LABORATORY 04 – APPLICATION LAYER & PHYSICAL LAYER PROTOCOLS

AUTORS

Andrés Felipe Chavarro Plazas

David Santiago Espinosa Rojas

TEACHER

Diego Alejandro Murcia Cespedes

ESCUELA COLOMBIANA DE INGENIERÍA JULIO GARAVITO

VIGILADA MINEDUCACIÓN

COMPUTER NETWORKS

GROUP 01

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OBJETIVE

The objective of this lab is to monitor application layer protocols, analyze their behavior, and understand their role in network communication. Additionally, it aims to review the structured cabling standard and its practical applications in network installations. Finally, the lab includes hands-on practice in performing cable punching with RJ-45 connectors and patch panels to ensure proper connectivity and compliance with industry standards.

MATERIALS & TOOLS

Some of the following materials are found in the networking laboratory and are provided by the university, others are brought by students for manual practice on cables.

- Laboratory computers with internet access
- Patch panel & face plate.
- Punch tools.
- Cable strippers & wire cutter.
- 4 to 6 meters of UTP/FTP CAT5 or CAT6.
- RJ-45 connectors.
- Cable tester.

INTRODUCTION

Modern Modern IT infrastructures depend on both physical and application layer technologies to ensure reliable connectivity, efficient communication, and secure data transmission. Organizations integrate wired and wireless networks to connect user devices, physical and virtualized servers, and cloud-based services through structured cabling and network protocols. Proper implementation of these elements is essential for maintaining a stable and scalable network.

In this lab, we will focus on two key aspects of networking: monitoring application layer protocols and understanding structured cabling. We will analyze network traffic using tools such as Wireshark, configure essential services like DNS, HTTP, and FTP, and examine how different application layer protocols interact with the transport layer. Additionally, we will apply structured cabling techniques, including cable crimping, patch panel setup, and connectivity testing to ensure proper network performance.

By combining theoretical concepts with hands-on experience, we aim to develop a deeper understanding of network infrastructure and management. This lab will not only reinforce our knowledge of application and physical layer interactions but also provide practical skills necessary for real-world networking environments.

THEORETICAL FRAMEWORK

The Application Layer is the topmost layer of the OSI model and is responsible for providing network services directly to end users. It includes protocols such as DNS, which translates domain names into IP addresses; HTTP, which enables web browsing; FTP, used for file transfers; and email protocols like SMTP and POP3, which handle email transmission and retrieval. In this lab, we will configure and analyze these protocols to understand their role in communication. The Transport Layer, while not the main focus, is essential for ensuring data reaches its destination reliably. It manages communication through ports, which identify specific services on a device. Protocols like TCP and UDP operate at this layer, with TCP ensuring reliable delivery and UDP providing faster but less reliable transmission. During the lab, we will observe how different ports correspond to application layer services. The Physical Layer deals with the hardware components of networking, including cables, connectors, and network devices. A key part of this lab is structured cabling, which follows standards like TIA/EIA-568 to ensure organized and scalable network installations.

In this lab, we will work with UTP cables, specifically CAT5 or CAT6, which are widely used for Ethernet networking. These cables consist of four twisted pairs of copper wires that reduce electromagnetic interference and crosstalk. CAT5, commonly used for Fast Ethernet. CAT5e, improves on CAT5, supporting Gigabit Ethernet with reduced crosstalk and improved transmission quality. CAT6 is designed for Gigabit Ethernet, with better shielding against interference. CAT6a, further enhances CAT6, making it ideal for high-speed networking and data centers. Each category uses RJ-45 connectors, and in this lab, we will perform cable crimping to create functional Ethernet cables. Additionally, we will configure and test a patch panel, which helps in managing cable connections in network infrastructure, ensuring proper organization, scalability, and troubleshooting efficiency.

To analyze network traffic, we will use Wireshark, a packet capture tool that allows us to inspect how data flows through the network. This tool captures packets at different layers of the OSI model, enabling us to observe protocol behavior, identify active application layer protocols, and troubleshoot communication issues. Wireshark is crucial for network diagnostics and security analysis, as it helps detect latency issues, misconfigurations, and potential attacks by analyzing real-time traffic patterns.

To complement this lab, we will use Cisco Packet Tracer, a network simulation tool that allows us to visualize, configure, and troubleshoot network topologies. This tool provides a controlled environment where devices such as routers, switches, servers, and workstations can be configured and tested. In this lab, Packet Tracer will allow us to configure a network with various devices, simulate the behavior of application layer protocols, analyze message flow, and use simulation mode to step through packet transmission. This will help us understand protocol interactions and troubleshoot potential issues before deploying configurations in real networks, making it an essential tool for network design and learning.

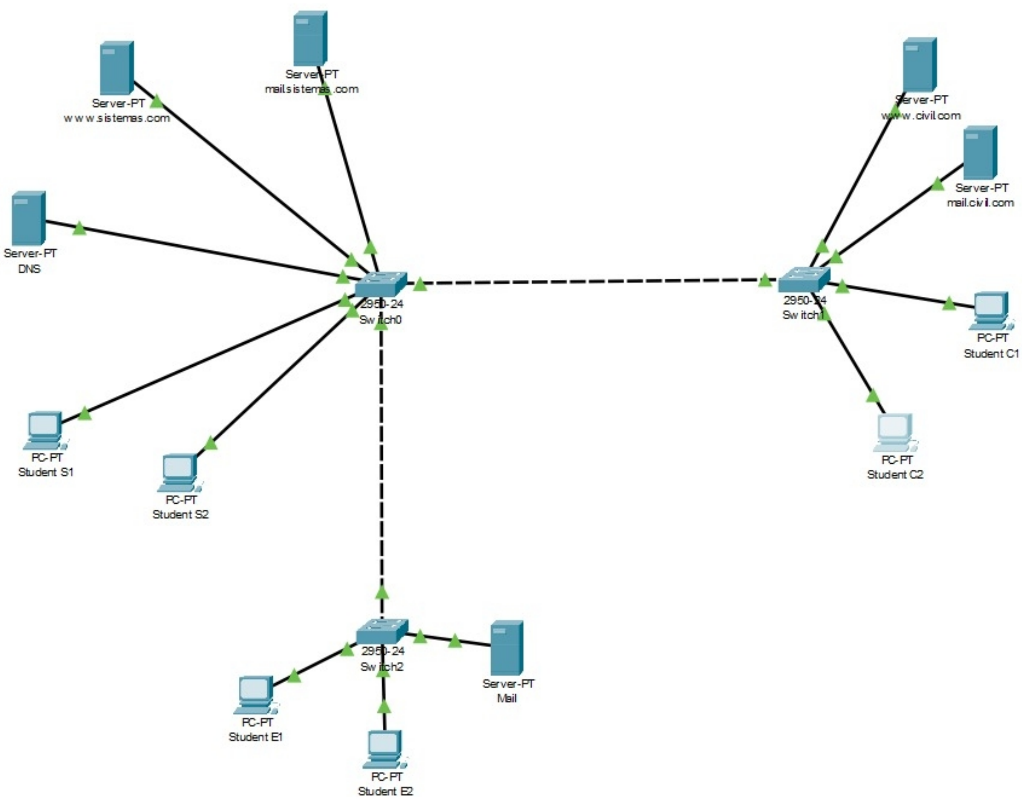
EXPERIMENTS

For this section, we will use software specialized in simulating network behavior, in this case Packet Tracer. In it, we will analyze the information obtained from the protocols and packets of the application and transport layers. In this way, we will be able to understand how these network tools really work, and we will be able to move on to a real environment.

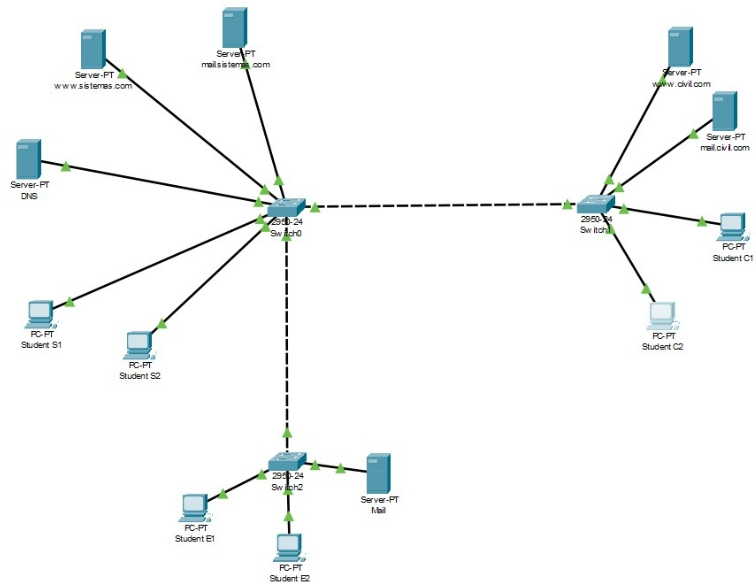
CISCO PACKET TRACER

In this first part, we will copy the given diagram and configure the different elements that we put in the Packet Tracer. In this way, we will be able to perform further operations at the following points.

- David’s diagram:

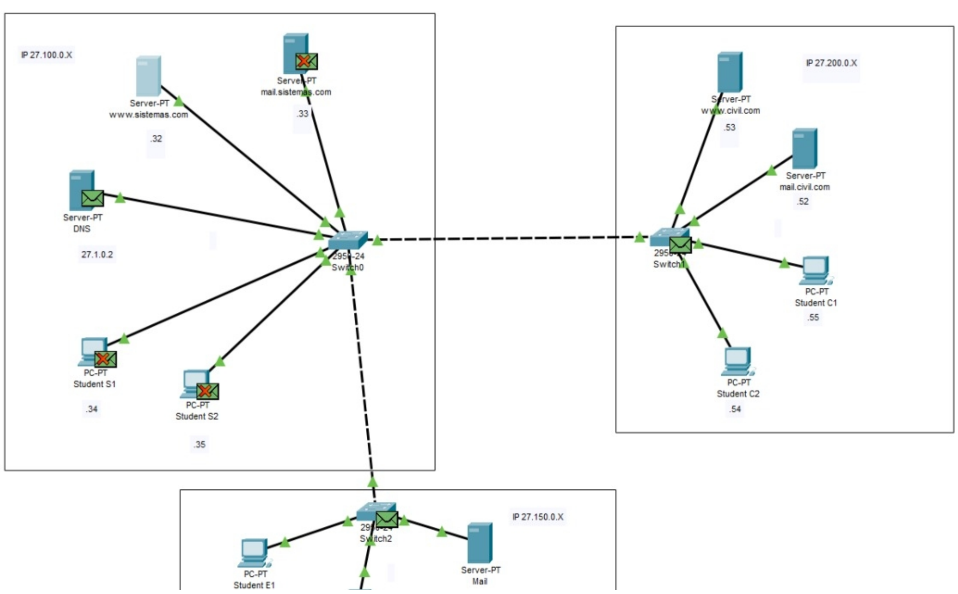


- Andres’s diagram:



NETWORK CONFIGURATION

Based on the diagram created, we will assign the characteristics that are given to us, such as names, IP's, DNS, and more configurations. Afterward, we will verify that the configuration is correct when we send a message with the corresponding tool, and it gives us a positive result.

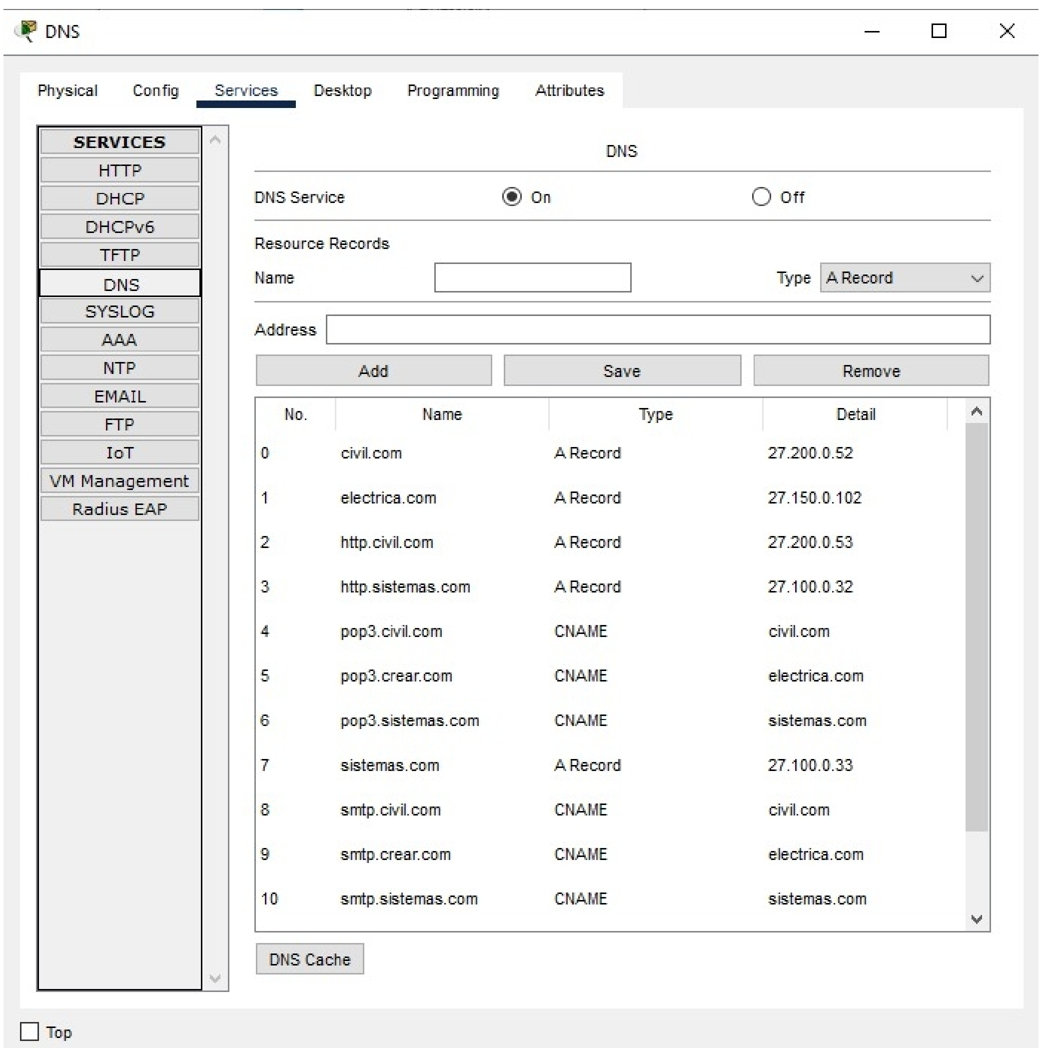


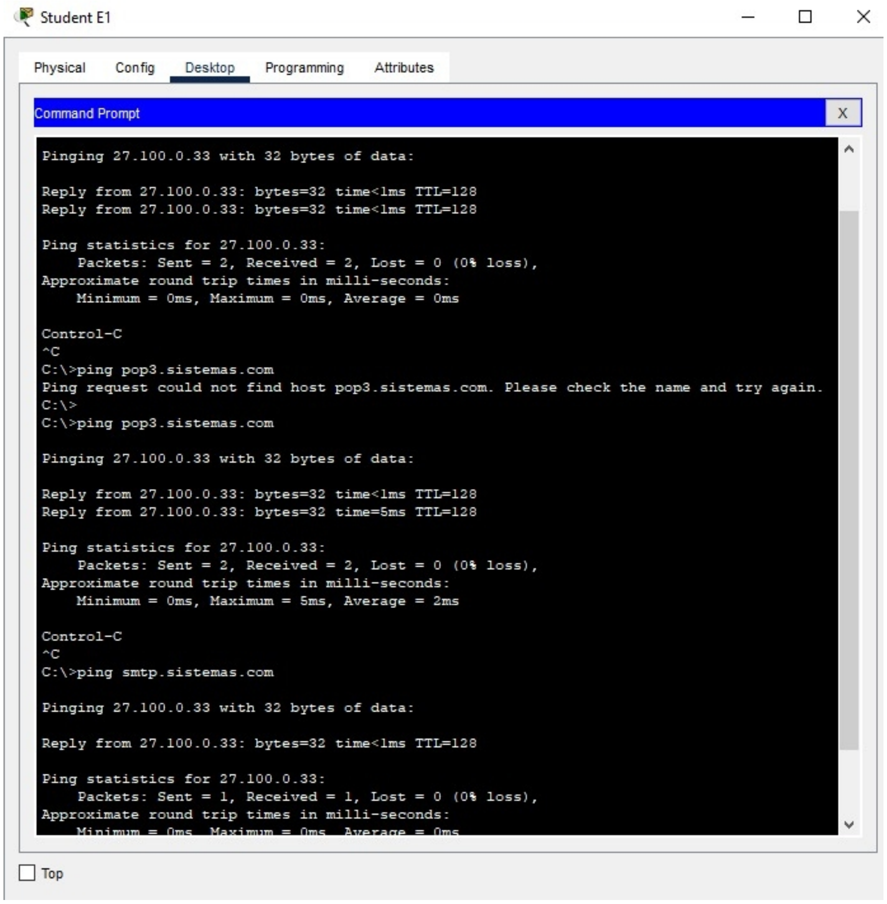
SERVICE CONFIGURATION

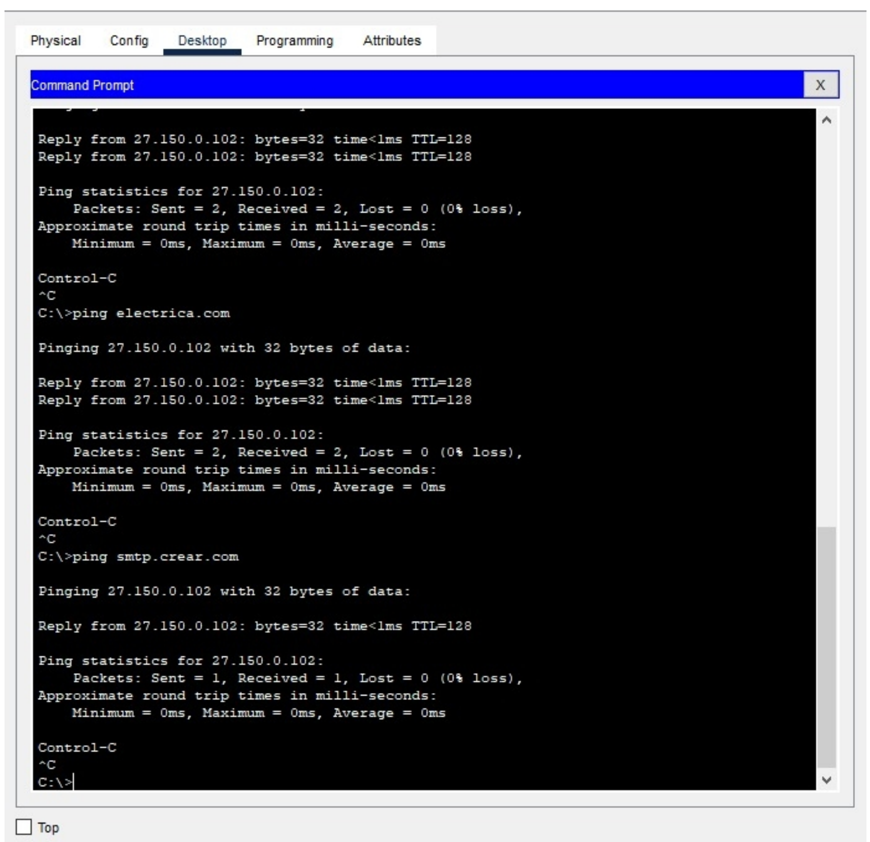
With the servers and devices already configured, we can move parameters within the services that each one has enabled. So, that we can check the functionality and data transmission between the different machines using application layer protocols, supported by those of the transport layer.

- **DNS**

We configured the requested DNS server with the properties and parameters that are in the guide, and then we tested them using the software's internal terminal, thus confirming that the connection is correct after starting the service.




Student E1
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Pinging 27.100.0.33 with 32 bytes of data:
Reply from 27.100.0.33: bytes=32 time<1ms TTL=128
Reply from 27.100.0.33: bytes=32 time<1ms TTL=128
Ping statistics for 27.100.0.33:
Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
^C
C:\>ping pop3.sistemas.com
Ping request could not find host pop3.sistemas.com. Please check the name and try again.
C:\>
C:\>ping pop3.sistemas.com
Pinging 27.100.0.33 with 32 bytes of data:
Reply from 27.100.0.33: bytes=32 time<1ms TTL=128
Reply from 27.100.0.33: bytes=32 time=5ms TTL=128
Ping statistics for 27.100.0.33:
Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 2ms
Control-C
^C
C:\>ping smtp.sistemas.com
Pinging 27.100.0.33 with 32 bytes of data:
Reply from 27.100.0.33: bytes=32 time<1ms TTL=128
Ping statistics for 27.100.0.33:
Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Top


Physical
Config
Desktop
Programming
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Command Prompt
X
Reply from 27.150.0.102: bytes=32 time<1ms TTL=128
Reply from 27.150.0.102: bytes=32 time<1ms TTL=128
Ping statistics for 27.150.0.102:
Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
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C:\>ping electrica.com
Pinging 27.150.0.102 with 32 bytes of data:
Reply from 27.150.0.102: bytes=32 time<1ms TTL=128
Reply from 27.150.0.102: bytes=32 time<1ms TTL=128
Ping statistics for 27.150.0.102:
Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
^C
C:\>ping smtp.crear.com
Pinging 27.150.0.102 with 32 bytes of data:
Reply from 27.150.0.102: bytes=32 time<1ms TTL=128
Ping statistics for 27.150.0.102:
Packets: Sent = 1, Received = 1, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
Control-C
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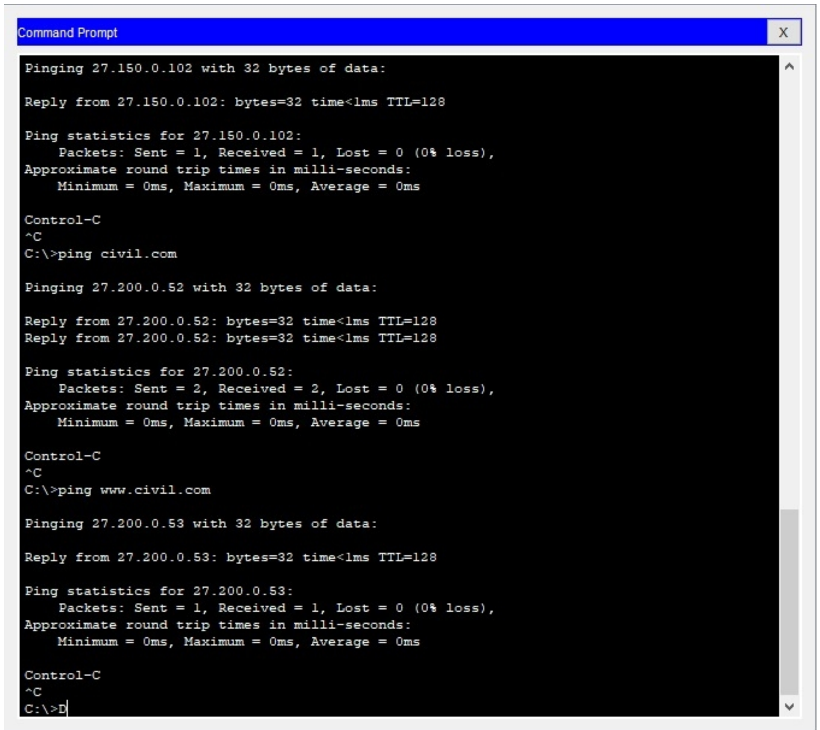
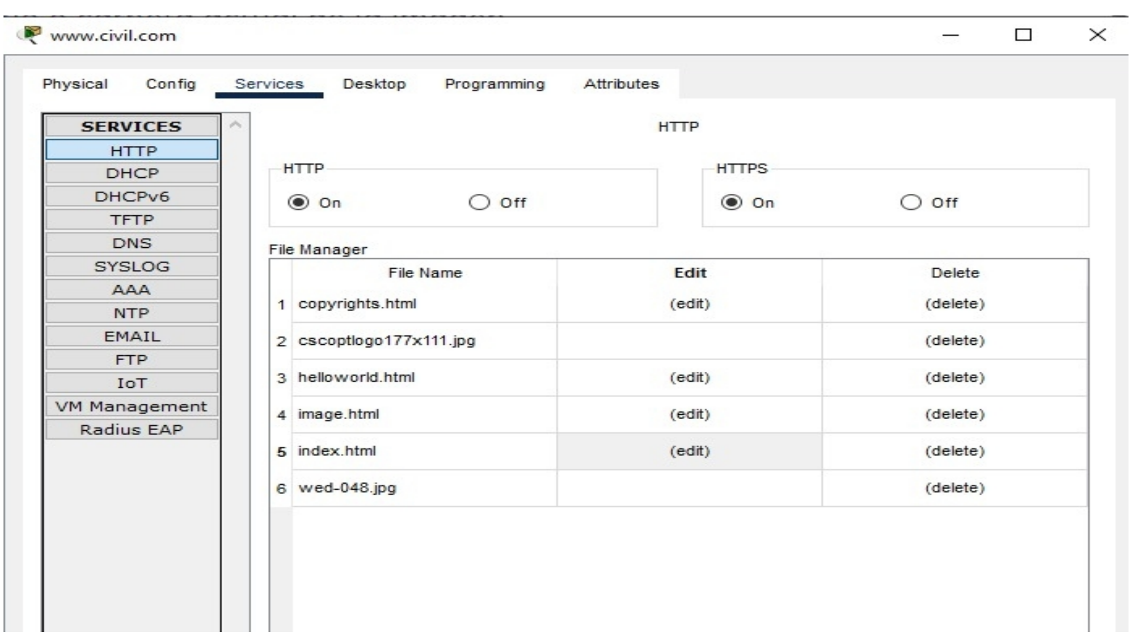
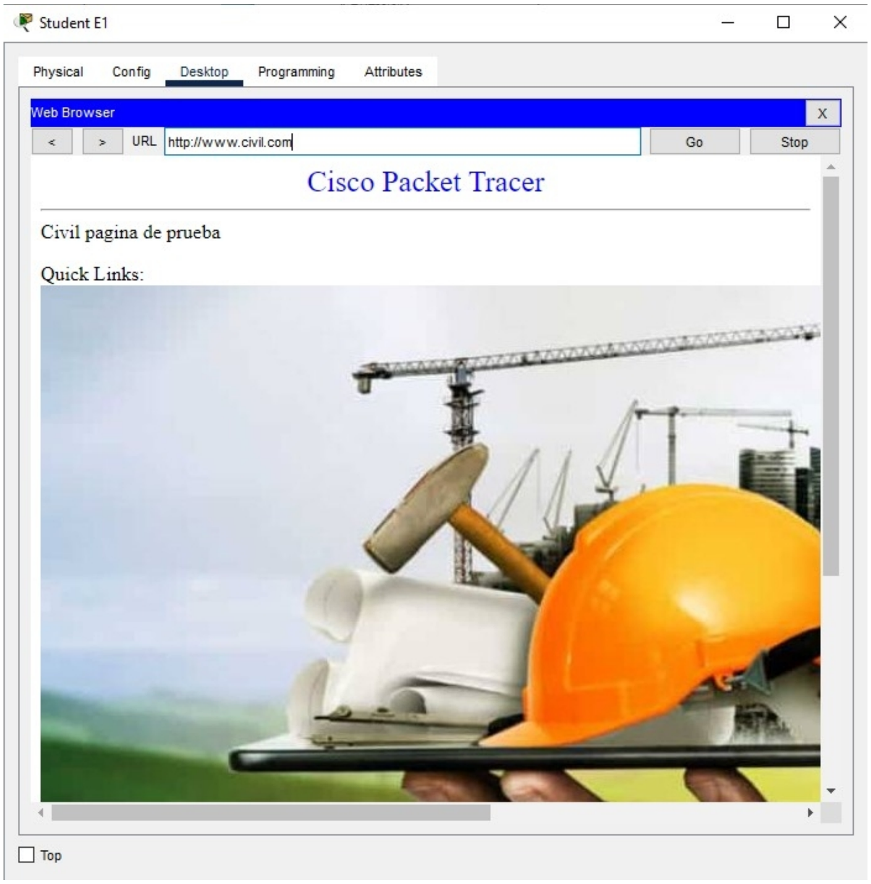
C:\>
Top



- **HTTP**

For web service servers, we are going to configure the HTTP service to interact with some web pages.




Student E1
Physical
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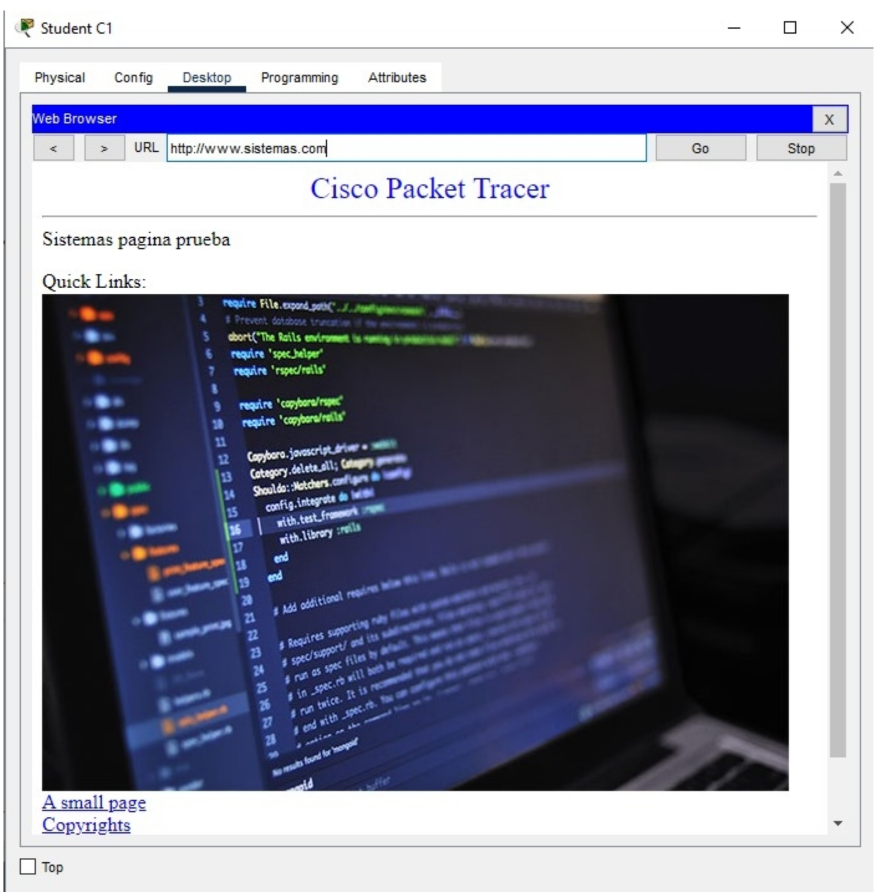
Student C1
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http://www.sistemas.com
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Cisco Packet Tracer
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Quick Links:
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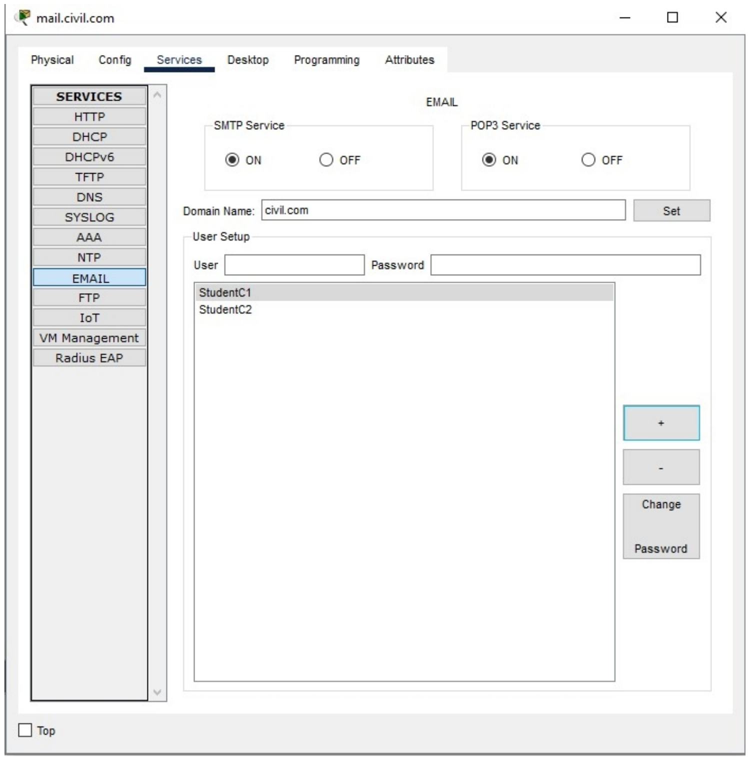
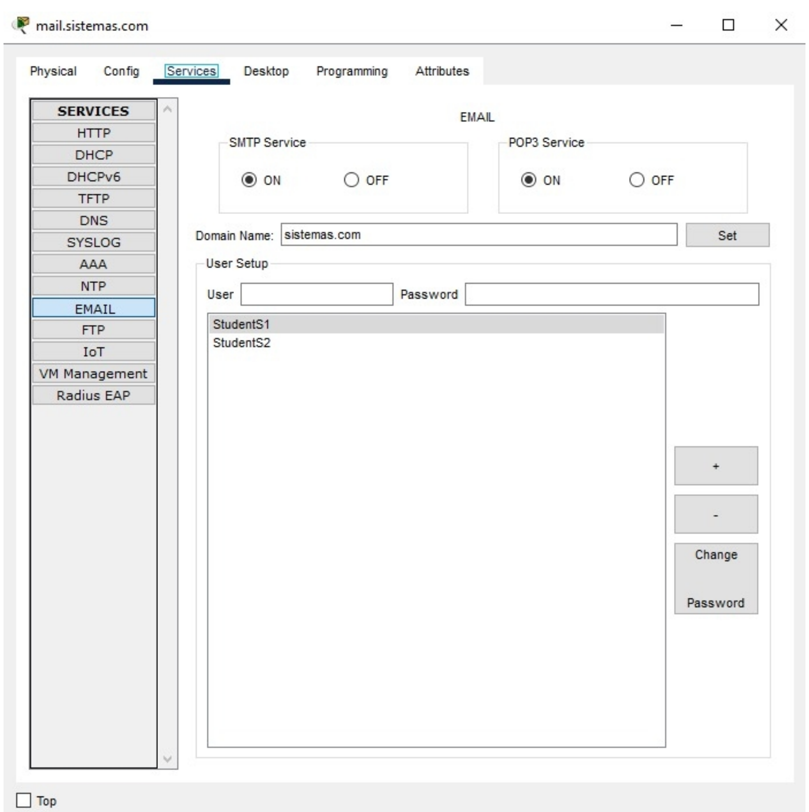
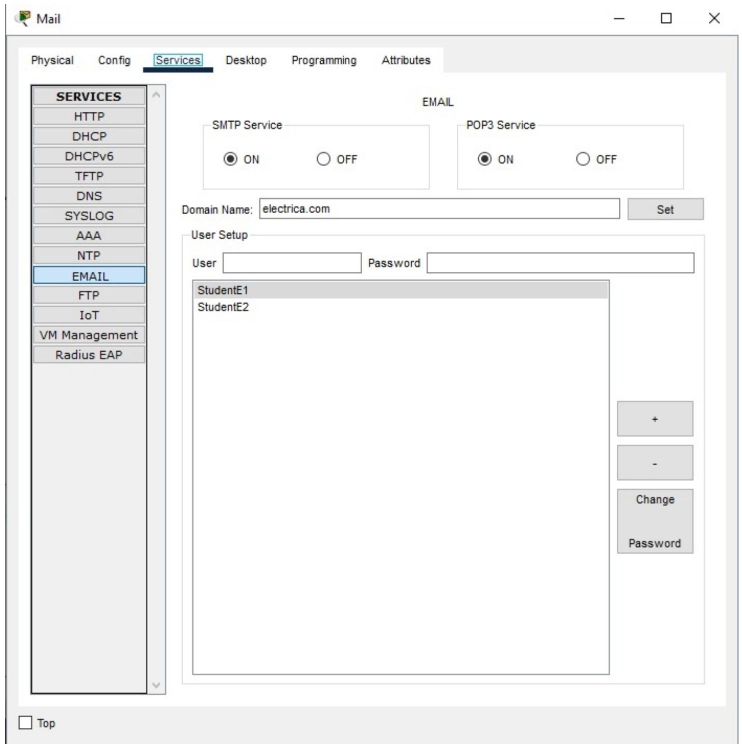
Top

- **MAIL**

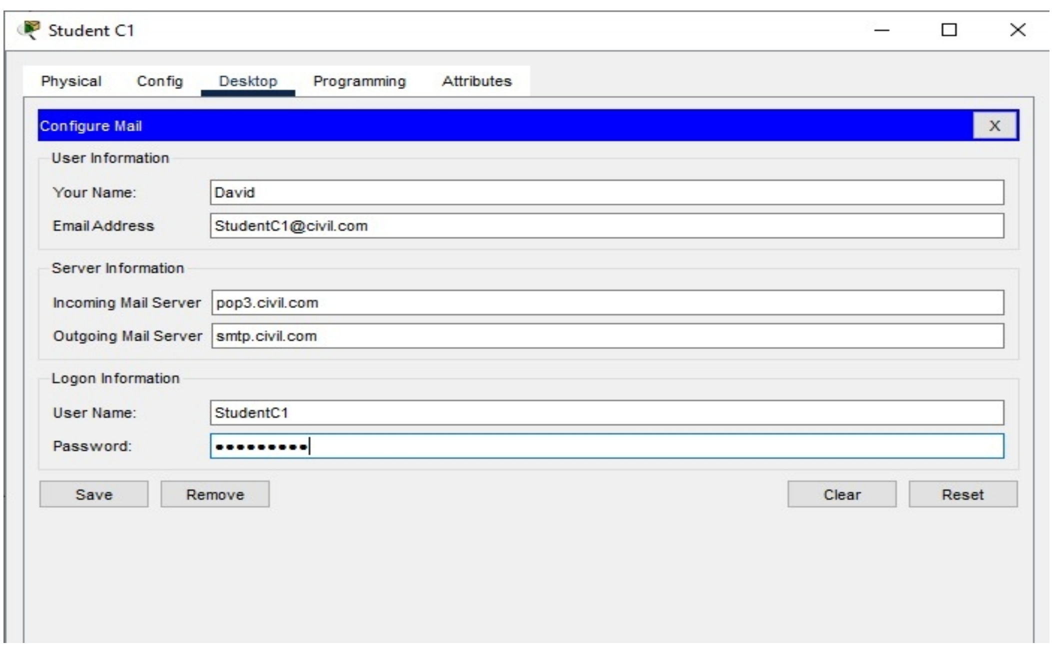
On the mail servers, we will configure the 3 faculties of the diagram with the settings set in the guide. Then, we will start the service to perform a couple of actions.

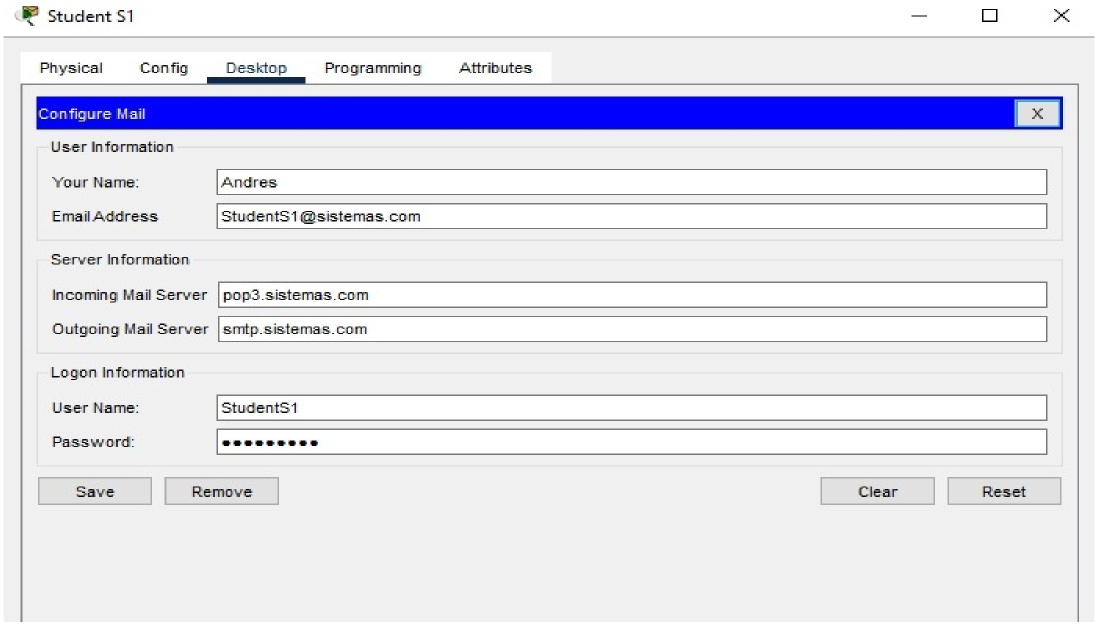




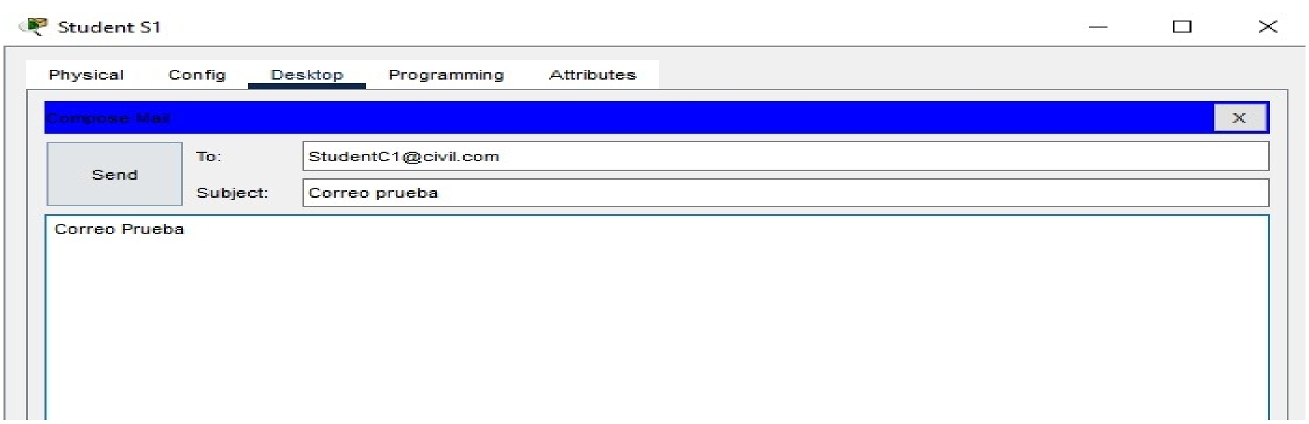


Now we entered the emails and send a test message. If successful, we will respond and verify that the system is fully configured.

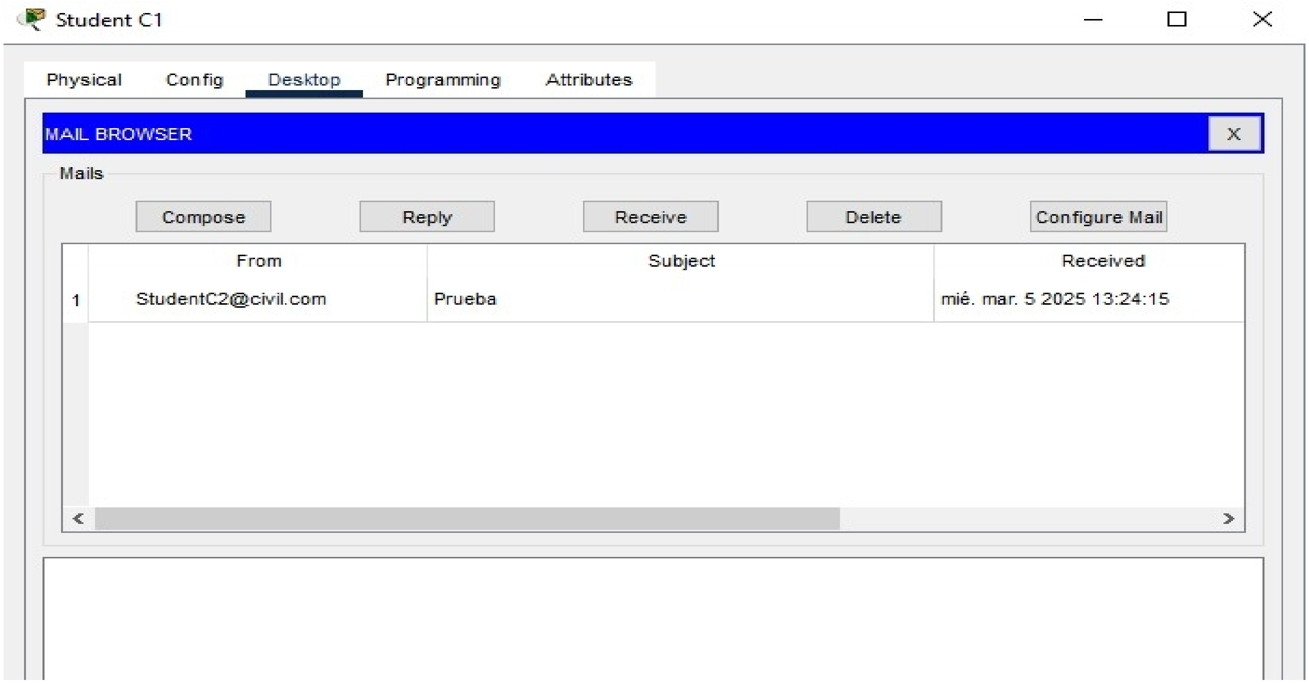


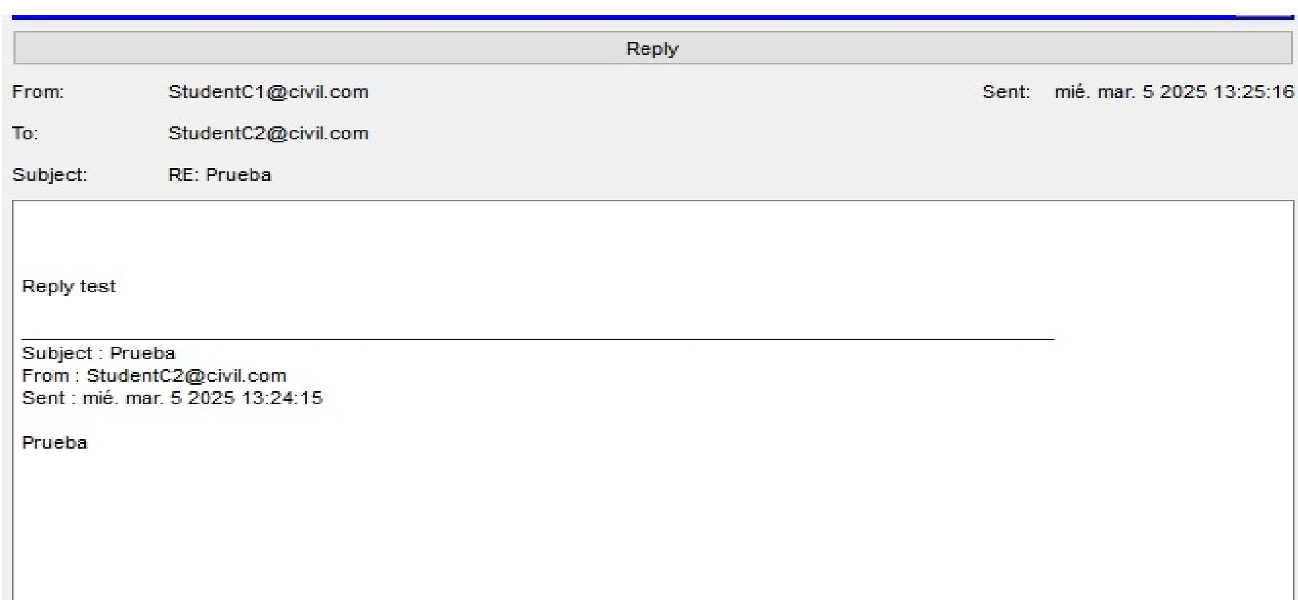


We send the message from one user to another.



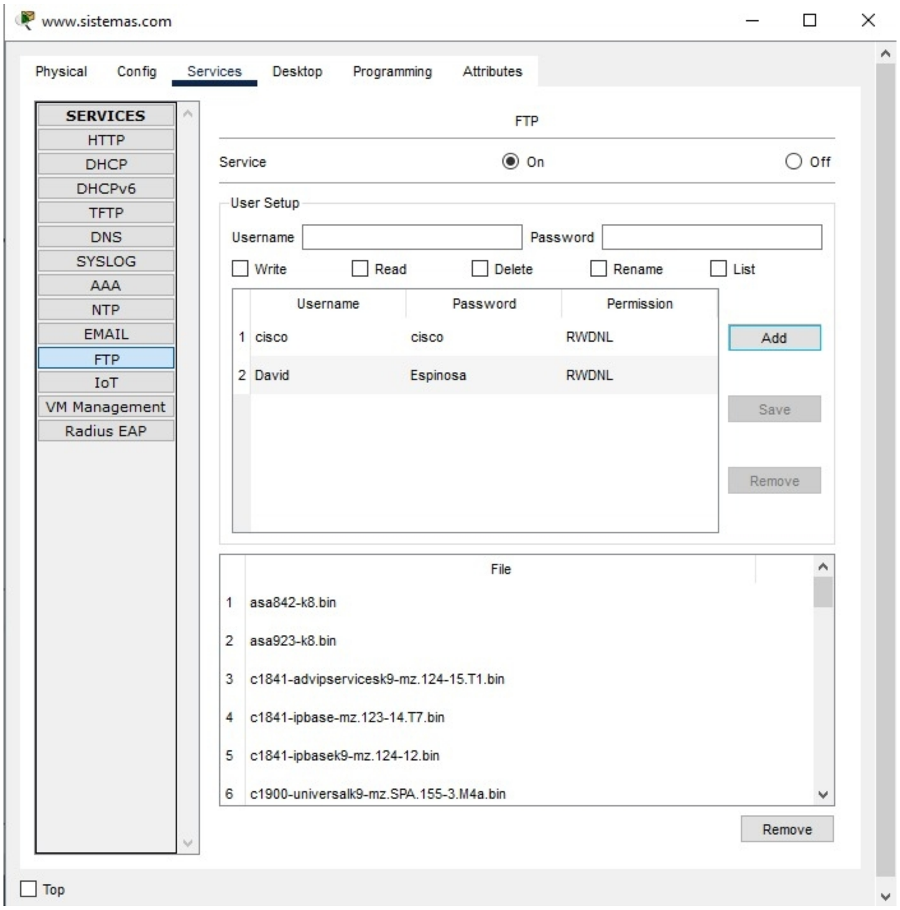
We respond and verify that the message reached the recipient.



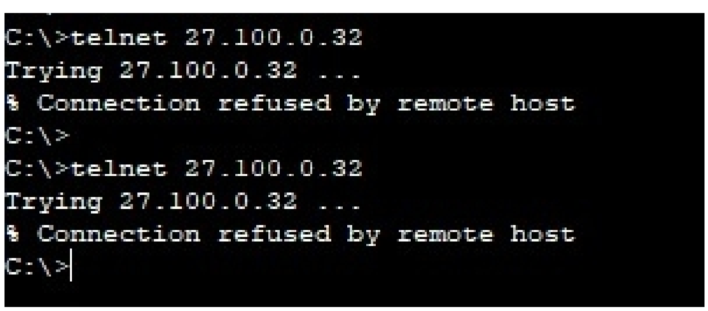


- **FTP**

For this protocol, we need to set up a user on a server with HTTP running, so that we can connect and perform the requested actions.



For the following, the `*telnet*` command was used but since it is not open from the switch on port 23 it does not allow the connection and download of files, added to this when using Telnet we would not be allowed to download files but only have remote access.

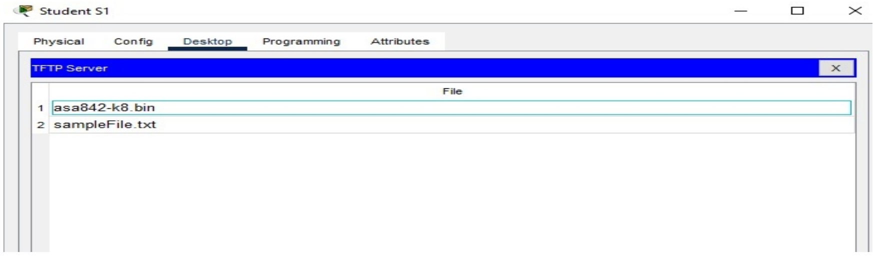


Then, we enter the credentials to obtain access to download the machine using the `***ftp***` command, which will allow us to perform remote access.

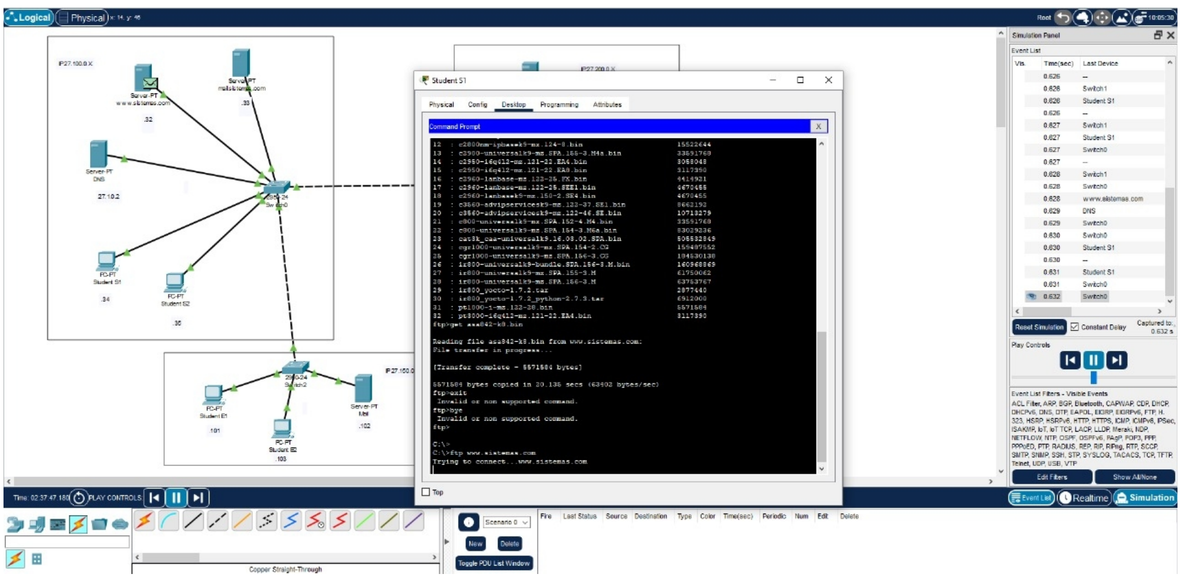
```
C:\>ftp www.sistemas.com
Trying to connect...www.sistemas.com
Connected to www.sistemas.com
220- Welcome to PT Ftp server
Username:David
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>
```

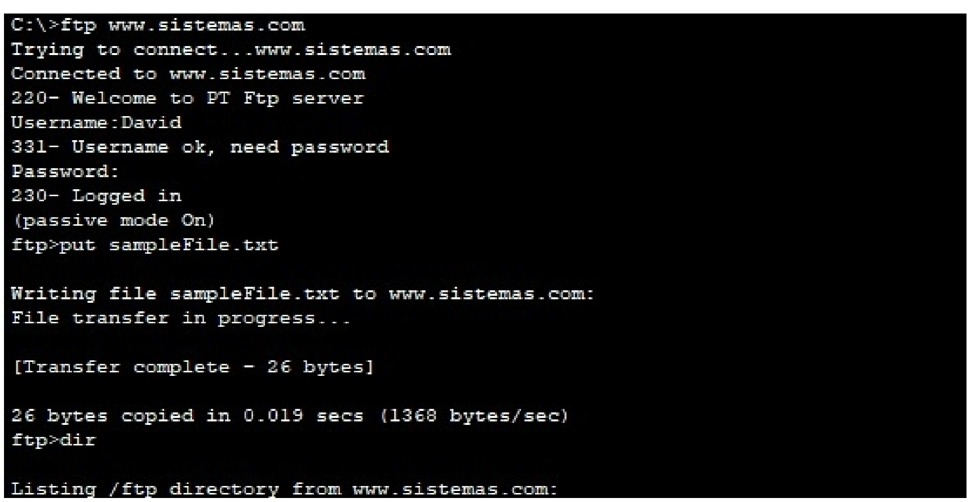
Now that we have access to the server, we are going to download the files we need, in this case a plain text file as an example. Also, we are going to enter the device tab and verify that it now has the files we downloaded through the protocol.

```
3    : c1841-ipbase-mz.123-14.T7.bin                      13832032
4    : c1841-ipbasek9-mz.124-12.bin                       16599160
5    : c1900-universalk9-mz.SPA.155-3.M4a.bin             33591768
6    : c2600-advipservicesk9-mz.124-15.T1.bin             33591768
7    : c2600-i-mz.122-28.bin                              5571584
8    : c2600-ipbasek9-mz.124-8.bin                        13169700
9    : c2800nm-advipservicesk9-mz.124-15.T1.bin           50938004
10   : c2800nm-advipservicesk9-mz.151-4.M4.bin            33591768
11   : c2800nm-ipbase-mz.123-14.T7.bin                    5571584
12   : c2800nm-ipbasek9-mz.124-8.bin                      15522644
13   : c2900-universalk9-mz.SPA.155-3.M4a.bin             33591768
14   : c2950-i6q4l2-mz.121-22.EA4.bin                     3058048
15   : c2950-i6q4l2-mz.121-22.EA8.bin                     3117390
16   : c2960-lanbase-mz.122-25.FX.bin                     4414921
17   : c2960-lanbase-mz.122-25.SEE1.bin                   4670455
18   : c2960-lanbasek9-mz.150-2.SE4.bin                   4670455
19   : c3560-advipservicesk9-mz.122-37.SE1.bin            8662192
20   : c3560-advipservicesk9-mz.122-46.SE.bin             10713279
21   : c800-universalk9-mz.SPA.152-4.M4.bin               33591768
22   : c800-universalk9-mz.SPA.154-3.M6a.bin              83029236
23   : cat3k_caa-universalk9.16.03.02.SPA.bin             505532849
24   : cgr1000-universalk9-mz.SPA.154-2.CG                159487552
25   : cgr1000-universalk9-mz.SPA.156-3.CG                184530138
26   : ir800-universalk9-bundle.SPA.156-3.M.bin           160968869
27   : ir800-universalk9-mz.SPA.155-3.M                   61750062
28   : ir800-universalk9-mz.SPA.156-3.M                   63753767
29   : ir800_yocto-1.7.2.tar                              2877440
30   : ir800_yocto-1.7.2_python-2.7.3.tar                 6912000
31   : pt1000-i-mz.122-28.bin                             5571584
32   : pt3000-i6q4l2-mz.121-22.EA4.bin                    3117390
ftp>get asa842-k8.bin

Reading file asa842-k8.bin from www.sistemas.com:
File transfer in progress...

[Transfer complete - 5571584 bytes]

5571584 bytes copied in 20.135 secs (63402 bytes/sec)
ftp>
```



Now that we know that this server works perfectly, we are going to carry out a final acceptance test using the simulation mode provided by Packet Trace. Thus, the system performs the same steps as previously but autonomously, resulting in no problems and downloading what had to be transferred.





```
C:\>ftp www.sistemas.com
Trying to connect...www.sistemas.com
Connected to www.sistemas.com
220- Welcome to PT Ftp server
Username:David
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>put sampleFile.txt

Writing file sampleFile.txt to www.sistemas.com:
File transfer in progress...

[Transfer complete - 26 bytes]

26 bytes copied in 0.019 secs (1368 bytes/sec)
ftp>dir

Listing /ftp directory from www.sistemas.com:
```

```
33   : sampleFile.txt                                   26
```

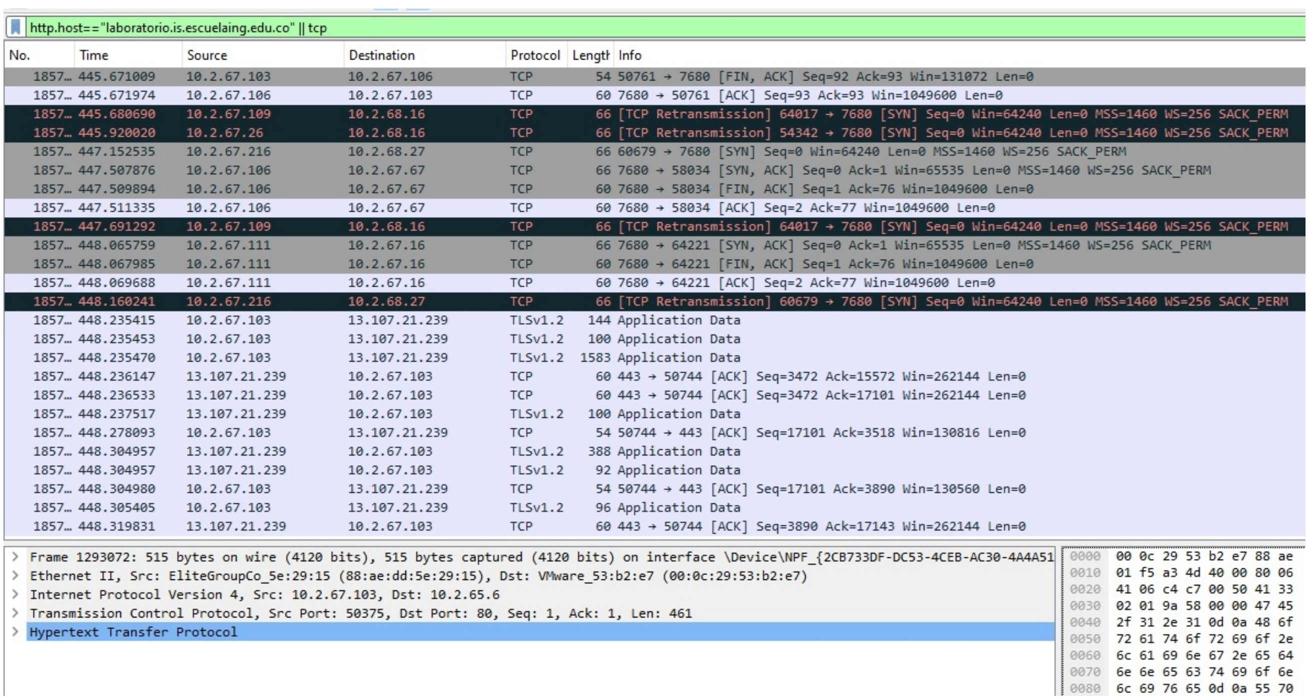
IN THE REAL NETWORK

With the experience gained in the simulator, we will apply the same concepts and practices in real environments and working with real hardware and networks.

WIRESHARK

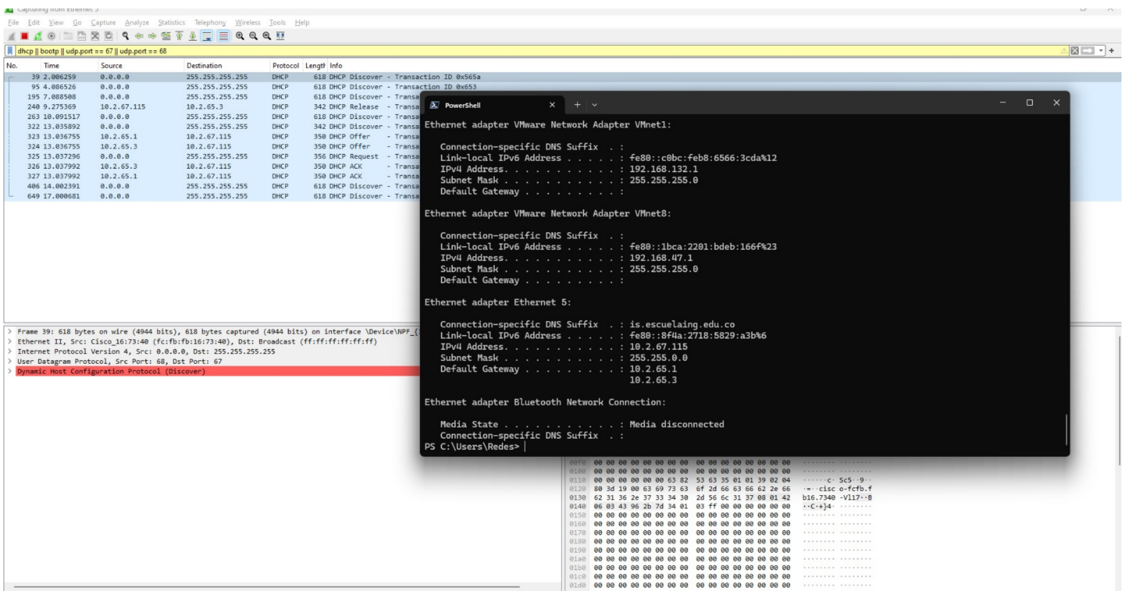
In this first part, we will use Wireshark to analyze how the application layer interacts with the transport layer through its ports. In this part, we will see how packets behave by analyzing each type found with queries.

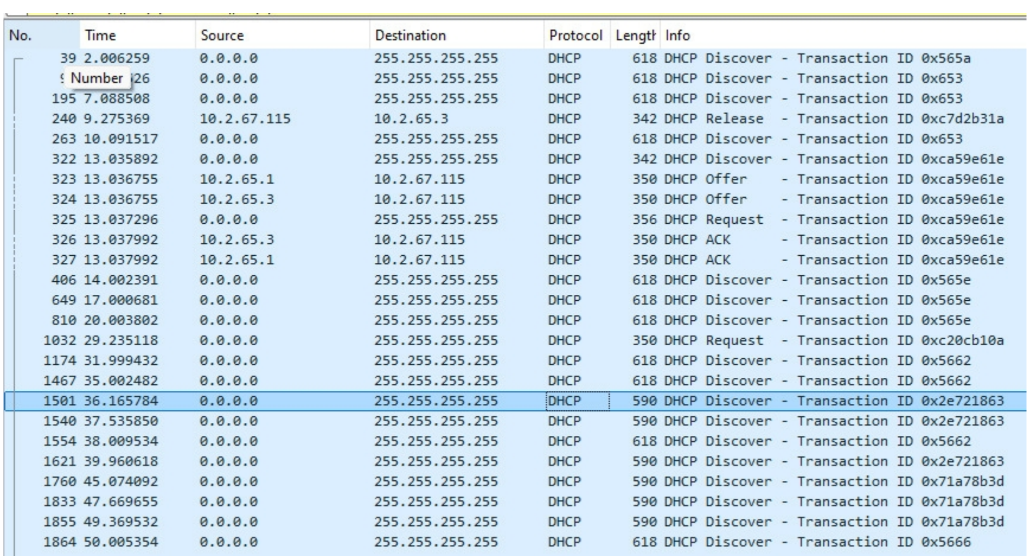
First, we are going to query the page that was given to us where some protocols of the target layers are shown. To do this, we place the filter `***http.host == "URL" && (tcp || dns || udp || http || tls || dns || ftp || ssh || telnet || smtp || pop || ntp || dhcp || smb )***`, which aims to capture the packets coming from the page on different ports.



Next, we are asked to query the packets coming from DHCP and the ports of the target layers. To do this, we perform the following query `***dhcp || bootp || udp.port == 67 || udp.port == 68***`, which mainly filters the known ports for connection to the network, all after requesting that the DHCP protocol be running again with the command `***ipconfig /release ipconfig /renew***`, after that, some packages will apear. There are some types:

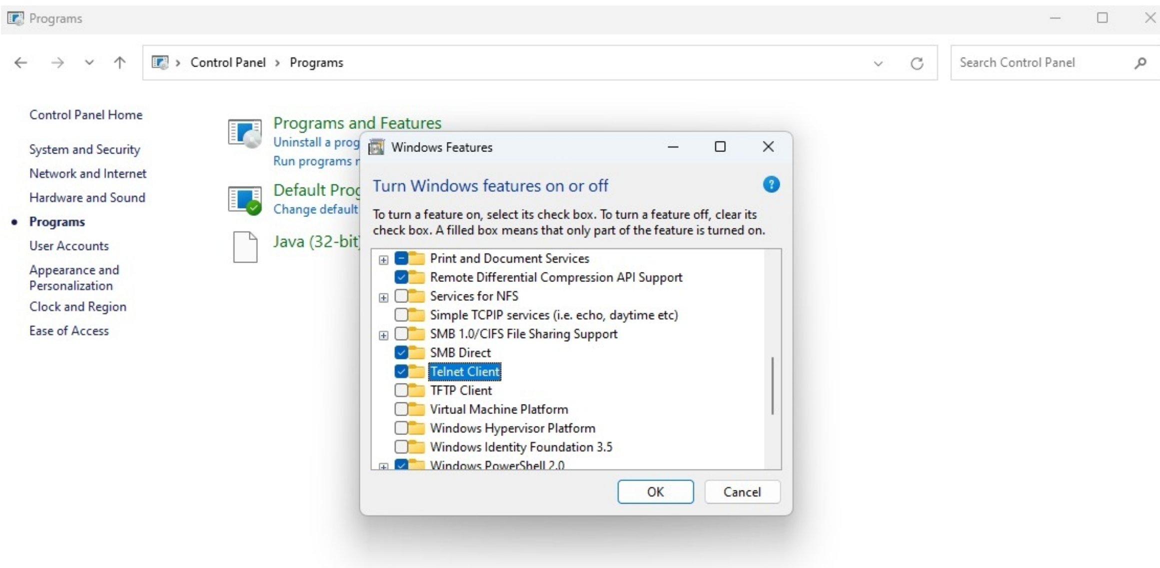
- Discover request to obtain IP.
- Offer response from the DHCP server.
- Request: confirmation from the client.
- ACK final confirmation from the server.



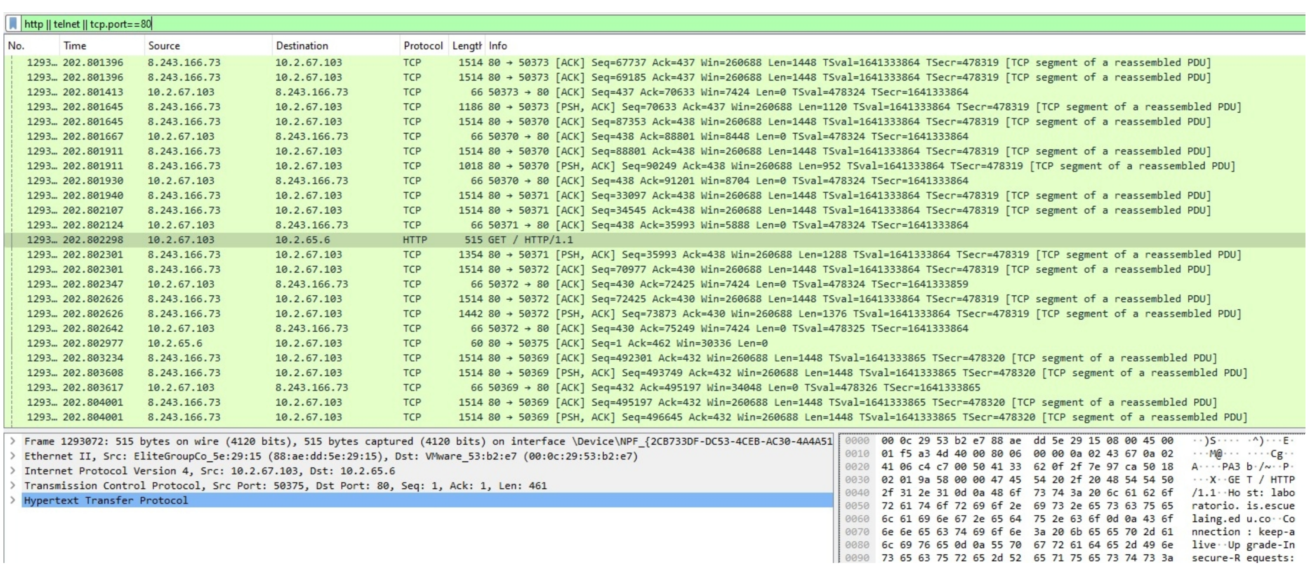


No.	Time	Source	Destination	Protocol	Length	Info
39	2.006259	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x565a
[illegible]	[illegible]26	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x653
195	7.088508	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x653
240	9.275369	10.2.67.115	10.2.65.3	DHCP	342	DHCP Release - Transaction ID 0xc7d2b31a
263	10.091517	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x653
322	13.035892	0.0.0.0	255.255.255.255	DHCP	342	DHCP Discover - Transaction ID 0xca59e61e
323	13.036755	10.2.65.1	10.2.67.115	DHCP	350	DHCP Offer - Transaction ID 0xca59e61e
324	13.036755	10.2.65.3	10.2.67.115	DHCP	350	DHCP Offer - Transaction ID 0xca59e61e
325	13.037296	0.0.0.0	255.255.255.255	DHCP	356	DHCP Request - Transaction ID 0xca59e61e
326	13.037992	10.2.65.3	10.2.67.115	DHCP	350	DHCP ACK - Transaction ID 0xca59e61e
327	13.037992	10.2.65.1	10.2.67.115	DHCP	350	DHCP ACK - Transaction ID 0xca59e61e
406	14.002391	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x565e
649	17.000681	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x565e
810	20.003802	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x565e
1032	29.235118	0.0.0.0	255.255.255.255	DHCP	350	DHCP Request - Transaction ID 0xc20cb10a
1174	31.999432	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x5662
1467	35.002482	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x5662
1501	36.165784	0.0.0.0	255.255.255.255	DHCP	590	DHCP Discover - Transaction ID 0x2e721863
1540	37.535850	0.0.0.0	255.255.255.255	DHCP	590	DHCP Discover - Transaction ID 0x2e721863
1554	38.009534	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x5662
1621	39.960618	0.0.0.0	255.255.255.255	DHCP	590	DHCP Discover - Transaction ID 0x2e721863
1760	45.074092	0.0.0.0	255.255.255.255	DHCP	590	DHCP Discover - Transaction ID 0x71a78b3d
1833	47.669655	0.0.0.0	255.255.255.255	DHCP	590	DHCP Discover - Transaction ID 0x71a78b3d
1855	49.369532	0.0.0.0	255.255.255.255	DHCP	590	DHCP Discover - Transaction ID 0x71a78b3d
1864	50.005354	0.0.0.0	255.255.255.255	DHCP	618	DHCP Discover - Transaction ID 0x5666

Finally, we are going to activate the Telnet protocol in order to capture the DHCP traffic. In our case, we enter the following part of the operating system; control panel>programs>Windows features. In this section, we have to activate the “Telnet Client” package, after which we can continue with the exercise.



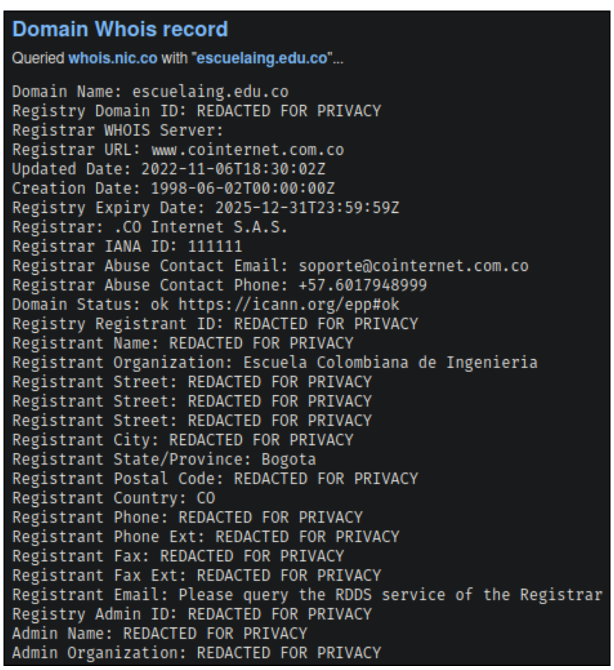
Using filters, we will place the following query `***tcp.port == 80 || tcp.port == 23***`, which will allow us to obtain Telnet and HTTP packets. This will allow us to make HTTP and TELNET requests.



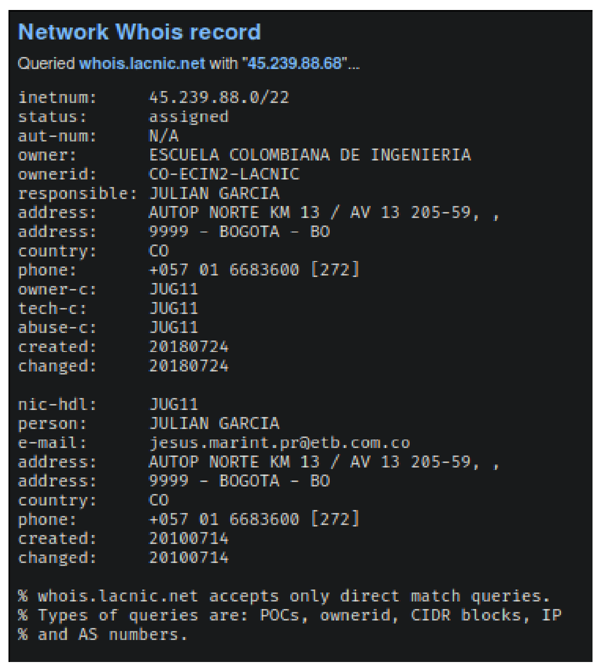
DNS SERVICE TEST

In this part, we have to do a test to examine and understand how DNS works in very specific cases. To do this, we perform a very specific test with the following 4 domains:

- **Escuelaing.edu.co**:
  - **Number of servers**: only 2 server: *ns1.escuelaing.edu.co* y *ns2.escuelaing.edu.co*.
  - **Life time**: since 02/05/1998 (+26 years).
  - **Person who registers the domain**: cointernet.
  - **Registration entity's ID**: 111111.
  - **Last update**: 06/10/2022 (+2 years).
  - **Record valid until**: 31/12/2025 (10 months remaining).
  - **Assigned IP range and by which registration authority**: 45.239.88.0/22 by LACNIC.
  - **Assigned company**: Escuela Colombiana de Ingeniería Julio Garavito.



```
Domain Whois record
Queried whois.nic.co with "escuelaing.edu.co"...

Domain Name: escuelaing.edu.co
Registry Domain ID: REDACTED FOR PRIVACY
Registrar WHOIS Server:
Registrar URL: www.cointernet.com.co
Updated Date: 2022-11-06T18:30:02Z
Creation Date: 1998-06-02T00:00:00Z
Registry Expiry Date: 2025-12-31T23:59:59Z
Registrar: .CO Internet S.A.S.
Registrar IANA ID: 111111
Registrar Abuse Contact Email: soporte@cointernet.com.co
Registrar Abuse Contact Phone: +57.6017948999
Domain Status: ok https://icann.org/epp#ok
Registry Registrant ID: REDACTED FOR PRIVACY
Registrant Name: REDACTED FOR PRIVACY
Registrant Organization: Escuela Colombiana de Ingenieria
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant City: REDACTED FOR PRIVACY
Registrant State/Province: Bogota
Registrant Postal Code: REDACTED FOR PRIVACY
Registrant Country: CO
Registrant Phone: REDACTED FOR PRIVACY
Registrant Phone Ext: REDACTED FOR PRIVACY
Registrant Fax: REDACTED FOR PRIVACY
Registrant Fax Ext: REDACTED FOR PRIVACY
Registrant Email: Please query the RDDS service of the Registrar
Registry Admin ID: REDACTED FOR PRIVACY
Admin Name: REDACTED FOR PRIVACY
Admin Organization: REDACTED FOR PRIVACY
```



```
Network Whois record
Queried whois.lacnic.net with "45.239.88.68"...

inetnum:     45.239.88.0/22
status:      assigned
aut-num:     N/A
owner:       ESCUELA COLOMBIANA DE INGENIERIA
ownerid:     CO-ECIN2-LACNIC
responsible: JULIAN GARCIA
address:     AUTOP NORTE KM 13 / AV 13 205-59, ,
address:     9999 - BOGOTA - BO
country:     CO
phone:       +057 01 6683600 [272]
owner-c:     JUG11
tech-c:      JUG11
abuse-c:     JUG11
created:     20180724
changed:     20180724

nic-hdl:     JUG11
person:      JULIAN GARCIA
e-mail:      jesus.marint.pr@etb.com.co
address:     AUTOP NORTE KM 13 / AV 13 205-59, ,
address:     9999 - BOGOTA - BO
country:     CO
phone:       +057 01 6683600 [272]
created:     20100714
changed:     20100714

% whois.lacnic.net accepts only direct match queries.
% Types of queries are: POCs, ownerid, CIDR blocks, IP
% and AS numbers.
```

- **jbb.gov.co**:
    - **Number of servers**: 2 servers; *ns31.domaincontrol.com* and *ns32.domaincontrol.com*.
    - **Life time**: since 20/01/2000 (+25 years).
    - **Person who registers the domain**: cointernet.
    - **Registration entity's ID**: Private for privacy.
    - **Last update**: 06/10/2022 (+2 years).
    - **Record valid until**: 20/01/2026 (10 months remaining).
    - **Assigned IP range and by which registration authority**: 20.33.0.0 – 20.128.255.255 from ARIN.
    - **Assigned company**: Microsoft Corporation.

Address lookup

canonical name	jbb.gov.co.
aliases	
addresses	20.94.123.146 20.119.228.39 2603:1030:403:3::a2

```
Domain Name: jbb.gov.co
Registry Domain ID: REDACTED FOR PRIVACY
Registrar WHOIS Server:
Registrar URL: www.cointernet.com.co
Updated Date: 2021-05-14T03:12:05Z
Creation Date: 2000-01-20T00:00:00Z
Registry Expiry Date: 2026-01-20T23:59:59Z
Registrar: .CO Internet S.A.S.
Registrar IANA ID: 111111
Registrar Abuse Contact Email: soporte@cointernet.com.co
Registrar Abuse Contact Phone: +57.6017948999
Domain Status: ok https://icann.org/epp#ok
Registry Registrant ID: REDACTED FOR PRIVACY
Registrant Name: REDACTED FOR PRIVACY
Registrant Organization: Jardin Bitanico Jose Celestino Mutis
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant Street: REDACTED FOR PRIVACY
Registrant City: REDACTED FOR PRIVACY
Registrant State/Province: Bogota
Registrant Postal Code: REDACTED FOR PRIVACY
Registrant Country: CO
Registrant Phone: REDACTED FOR PRIVACY
Registrant Phone Ext: REDACTED FOR PRIVACY
Registrant Fax: REDACTED FOR PRIVACY
Registrant Fax Ext: REDACTED FOR PRIVACY
Registrant Email: Please query the RDDS service of the Registrar
Registry Admin ID: REDACTED FOR PRIVACY
Admin Name: REDACTED FOR PRIVACY
Admin Organization: REDACTED FOR PRIVACY
Admin Street: REDACTED FOR PRIVACY
Admin Street: REDACTED FOR PRIVACY
Admin Street: REDACTED FOR PRIVACY
Admin City: REDACTED FOR PRIVACY
Admin State/Province: REDACTED FOR PRIVACY
Admin Postal Code: REDACTED FOR PRIVACY
Admin Country: REDACTED FOR PRIVACY
Admin Phone: REDACTED FOR PRIVACY
Admin Phone Ext: REDACTED FOR PRIVACY
Admin Fax: REDACTED FOR PRIVACY
Admin Fax Ext: REDACTED FOR PRIVACY
```

```
Admin Organization: REDACTED FOR PRIVACY
Admin Street: REDACTED FOR PRIVACY
Admin Street: REDACTED FOR PRIVACY
Admin Street: REDACTED FOR PRIVACY
Admin City: REDACTED FOR PRIVACY
Admin State/Province: REDACTED FOR PRIVACY
Admin Postal Code: REDACTED FOR PRIVACY
Admin Country: REDACTED FOR PRIVACY
Admin Phone: REDACTED FOR PRIVACY
Admin Phone Ext: REDACTED FOR PRIVACY
Admin Fax: REDACTED FOR PRIVACY
Admin Fax Ext: REDACTED FOR PRIVACY
Admin Email: Please query the RDDS service of the Registrar of Re
Registry Tech ID: REDACTED FOR PRIVACY
Tech Name: REDACTED FOR PRIVACY
Tech Organization: REDACTED FOR PRIVACY
Tech Street: REDACTED FOR PRIVACY
Tech Street: REDACTED FOR PRIVACY
Tech Street: REDACTED FOR PRIVACY
Tech City: REDACTED FOR PRIVACY
Tech State/Province: REDACTED FOR PRIVACY
Tech Postal Code: REDACTED FOR PRIVACY
Tech Country: REDACTED FOR PRIVACY
Tech Phone: REDACTED FOR PRIVACY
Tech Phone Ext: REDACTED FOR PRIVACY
Tech Fax: REDACTED FOR PRIVACY
Tech Fax Ext: REDACTED FOR PRIVACY
Tech Email: Please query the RDDS service of the Registrar of Rec
Name Server: ns31.domaincontrol.com
Name Server: ns32.domaincontrol.com
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.ica
>>> Last update of WHOIS database: 2025-03-08T05:29:04Z <<<
```

Network Whois record

Queried **whois.arin.net** with "**n 20.94.123.146**"...

```
NetRange:       20.33.0.0 - 20.128.255.255
CIDR:           20.48.0.0/12, 20.40.0.0/13, 20.33.0.0/16, 20.64.0.0/10, 20.128.0.0/16, 20.36.0.0/14, 20.34.0.0/15
NetName:        MSFT
NetHandle:      NET-20-33-0-0-1
Parent:         NET20 (NET-20-0-0-0-0)
NetType:        Direct Allocation
OriginAS:
Organization:   Microsoft Corporation (MSFT)
RegDate:        2017-10-18
Updated:        2021-12-14
Ref:            https://rdap.arin.net/registry/ip/20.33.0.0


OrgName:        Microsoft Corporation
OrgId:          MSFT
Address:        One Microsoft Way
City:           Redmond
StateProv:      WA
PostalCode:     98052
Country:        US
RegDate:        1998-07-10
Updated:        2024-03-18
Comment:        To report suspected security issues specific to traffic emanating from Microsoft online services,
Comment:        * https://cert.microsoft.com.
Comment:
Comment:        For SPAM and other abuse issues, such as Microsoft Accounts, please contact:
Comment:        * abuse@microsoft.com.
Comment:
Comment:        To report security vulnerabilities in Microsoft products and services, please contact:
Comment:        * secure@microsoft.com.
Comment:
Comment:        For legal and law enforcement-related requests, please contact:
Comment:        * msndcc@microsoft.com
Comment:
Comment:        For routing, peering or DNS issues, please
Comment:        contact:
Comment:        * IOC@microsoft.com
```

- **google.com**:
  - **Number of servers**: 6 Ipv4 and 4 Ipv6, total of 10 servers.
  - **Life time**: since 15/09/1997 (+27 years).
  - **Person who registers the domain**: MarkMonitor.
  - **Registration entity's ID**: 292.
  - **Last update**: 10/08/2024 (+1/2 year)
  - **Record valid until**: 13/09/2028 (+3 years remaining).
  - **Assigned IP range and by which registration authority**: 142.250.0.0 – 142.251.255.255 from ARIN.
  - **Assigned company**: Google LLC.

Address lookup

canonical name	google.com.
aliases	
addresses	142.250.114.138 142.250.114.100 142.250.114.139 142.250.114.102 142.250.114.101 142.250.114.113 2607:f8b0:4023:1009::66 2607:f8b0:4023:1009::64 2607:f8b0:4023:1009::65 2607:f8b0:4023:1009::71

Domain Whois record

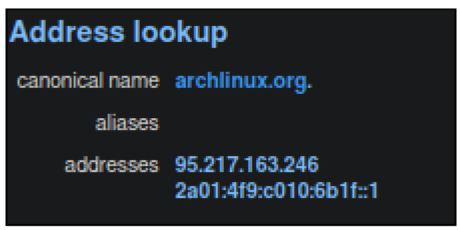
Queried whois.internic.net with "dom google.com"...

```
   Domain Name: GOOGLE.COM
   Registry Domain ID: 2138514_DOMAIN_COM-VRSN
   Registrar WHOIS Server: whois.markmonitor.com
   Registrar URL: http://www.markmonitor.com
   Updated Date: 2019-09-09T15:39:04Z
   Creation Date: 1997-09-15T04:00:00Z
   Registry Expiry Date: 2028-09-14T04:00:00Z
   Registrar: MarkMonitor Inc.
   Registrar IANA ID: 292
   Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
   Registrar Abuse Contact Phone: +1.2086851750
   Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
   Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
   Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
   Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
   Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
   Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
   Name Server: NS1.GOOGLE.COM
   Name Server: NS2.GOOGLE.COM
   Name Server: NS3.GOOGLE.COM
   Name Server: NS4.GOOGLE.COM
   DNSSEC: unsigned
   URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2025-03-07T17:00:34Z <<<
```

```
Domain Name: google.com
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2024-08-02T02:17:33+0000
Creation Date: 1997-09-15T07:00:00+0000
Registrar Registration Expiration Date: 2028-09-13T07:00:00+0000
Registrar: MarkMonitor, Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientUpdateProhibited (https://www.icann.org/epp#clientUpdateProhibited)
Domain Status: clientTransferProhibited (https://www.icann.org/epp#clientTransferProhibited)
Domain Status: clientDeleteProhibited (https://www.icann.org/epp#clientDeleteProhibited)
Domain Status: serverUpdateProhibited (https://www.icann.org/epp#serverUpdateProhibited)
Domain Status: serverTransferProhibited (https://www.icann.org/epp#serverTransferProhibited)
Domain Status: serverDeleteProhibited (https://www.icann.org/epp#serverDeleteProhibited)
Registrant Organization: Google LLC
Registrant State/Province: CA
Registrant Country: US
Registrant Email: Select Request Email Form at https://domains.markmonitor.com/whois/google.com
Admin Organization: Google LLC
Admin State/Province: CA
Admin Country: US
Admin Email: Select Request Email Form at https://domains.markmonitor.com/whois/google.com
Tech Organization: Google LLC
Tech State/Province: CA
Tech Country: US
Tech Email: Select Request Email Form at https://domains.markmonitor.com/whois/google.com
Name Server: ns2.google.com
Name Server: ns4.google.com
Name Server: ns1.google.com
Name Server: ns3.google.com
DNSSEC: unsigned
URL of the ICANN WHOIS Data Problem Reporting System: http://wdprs.internic.net/
>>> Last update of WHOIS database: 2025-03-07T16:59:31+0000 <<<
```

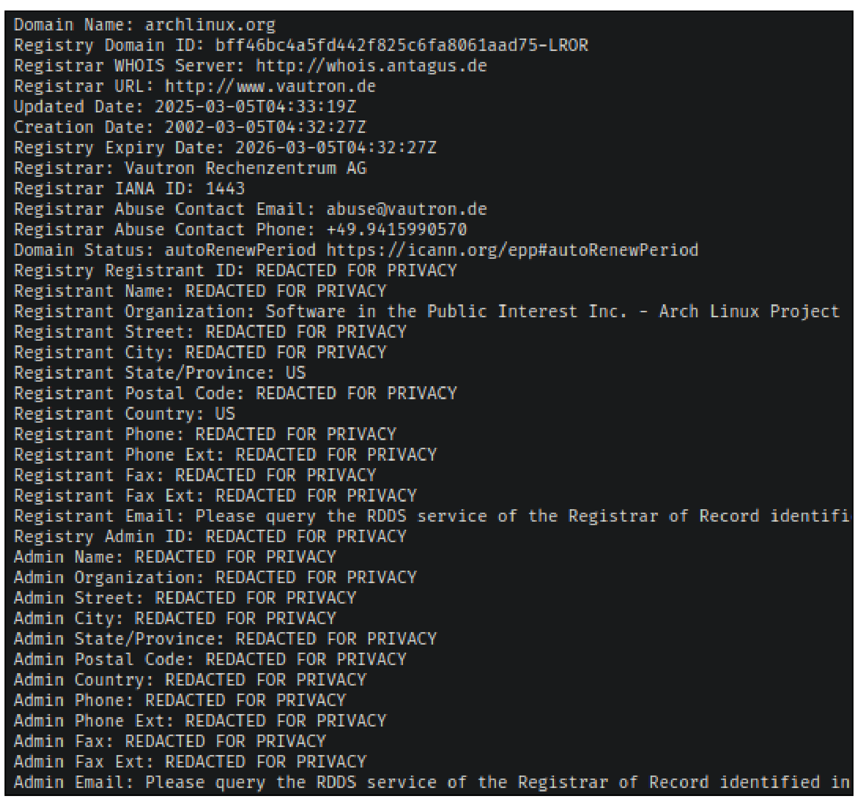
```
NetRange:       142.250.0.0 - 142.251.255.255
CIDR:           142.250.0.0/15
NetName:        GOOGLE
NetHandle:      NET-142-250-0-0-1
Parent:         NET142 (NET-142-0-0-0-0)
NetType:        Direct Allocation
OriginAS:       AS15169
Organization:   Google LLC (GOGL)
RegDate:        2012-05-24
Updated:        2012-05-24
Ref:            https://rdap.arin.net/registry/ip/142.250.0.0


OrgName:        Google LLC
OrgId:          GOGL
Address:        1600 Amphitheatre Parkway
City:           Mountain View
StateProv:      CA
PostalCode:     94043
Country:        US
RegDate:        2000-03-30
Updated:        2019-10-31
Comment:        Please note that the recommended way to file abuse complaints are located in the following links.
Comment:
Comment:        To report abuse and illegal activity: https://www.google.com/contact/
Comment:
Comment:        For legal requests: http://support.google.com/legal
Comment:
Comment:        Regards,
Comment:        The Google Team
Ref:            https://rdap.arin.net/registry/entity/GOGL
```

- **Arclinux.org**:
    - **Number of servers**: only 3 servers; *helium.ns.hetzner.de*, *hydrogen.ns.hetzner.com, oxygen.ns.hetzner.com*.
    - **Life time**: 05/03/2002 (+23 year).
    - **Person who registers the domain**: Vautron Rechenzentrum AG.
    - **Registration entity's ID**: bff46bc4a5fd442f825c6fa8061aad75-LROR.
    - **Last update**: 05/03/2025 (this month)
    - **Record valid until**: 05/03/2026 (1 year remaining).
    - **Assigned IP range and by which registration authority**: 95.217.160.0 – 95.217.167.255 from RIPE.
    - **Assigned company**: Hetzner Online GmbH (Germany).



Address lookup

canonical name	archlinux.org.
aliases	
addresses	95.217.163.246 2a01:4f9:c010:6b1f::1



```
Domain Name: archlinux.org
Registry Domain ID: bff46bc4a5fd442f825c6fa8061aad75-LROR
Registrar WHOIS Server: http://whois.antagus.de
Registrar URL: http://www.vautron.de
Updated Date: 2025-03-05T04:33:19Z
Creation Date: 2002-03-05T04:32:27Z
Registry Expiry Date: 2026-03-05T04:32:27Z
Registrar: Vautron Rechenzentrum AG
Registrar IANA ID: 1443
Registrar Abuse Contact Email: abuse@vautron.de
Registrar Abuse Contact Phone: +49.9415990570
Domain Status: autoRenewPeriod https://icann.org/epp#autoRenewPeriod
Registry Registrant ID: REDACTED FOR PRIVACY
Registrant Name: REDACTED FOR PRIVACY
Registrant Organization: Software in the Public Interest Inc. - Arch Linux Project
Registrant Street: REDACTED FOR PRIVACY
Registrant City: REDACTED FOR PRIVACY
Registrant State/Province: US
Registrant Postal Code: REDACTED FOR PRIVACY
Registrant Country: US
Registrant Phone: REDACTED FOR PRIVACY
Registrant Phone Ext: REDACTED FOR PRIVACY
Registrant Fax: REDACTED FOR PRIVACY
Registrant Fax Ext: REDACTED FOR PRIVACY
Registrant Email: Please query the RDDS service of the Registrar of Record identifi
Registry Admin ID: REDACTED FOR PRIVACY
Admin Name: REDACTED FOR PRIVACY
Admin Organization: REDACTED FOR PRIVACY
Admin Street: REDACTED FOR PRIVACY
Admin City: REDACTED FOR PRIVACY
Admin State/Province: REDACTED FOR PRIVACY
Admin Postal Code: REDACTED FOR PRIVACY
Admin Country: REDACTED FOR PRIVACY
Admin Phone: REDACTED FOR PRIVACY
Admin Phone Ext: REDACTED FOR PRIVACY
Admin Fax: REDACTED FOR PRIVACY
Admin Fax Ext: REDACTED FOR PRIVACY
Admin Email: Please query the RDDS service of the Registrar of Record identified in
```

Network Whois record

Queried **whois.ripe.net** with "**-B 95.217.163.246**"...

```
% Information related to '95.217.160.0 - 95.217.167.255'

% Abuse contact for '95.217.160.0 - 95.217.167.255' is 'abuse@hetzner.com'

inetnum:        95.217.160.0 - 95.217.167.255
netname:        CLOUD-HEL1
country:        FI
status:         ASSIGNED PA
org:            ORG-HOA1-RIPE
admin-c:        HOAC1-RIPE
tech-c:         HOAC1-RIPE
mnt-by:         HOS-GUN
remarks:        INFRA-AW
created:        2023-12-12T12:40:45Z
last-modified:  2023-12-12T12:40:45Z
source:         RIPE

organisation:   ORG-HOA1-RIPE
org-name:       Hetzner Online GmbH
country:        DE
org-type:       LIR
address:        Industriestrasse 25
address:        D-91710
address:        Gunzenhausen
address:        GERMANY
phone:          +49 9831 5050
fax-no:         +49 9831 5053
e-mail:         info@hetzner.de
admin-c:        MF1400-RIPE
admin-c:        GM834-RIPE
admin-c:        HOAC1-RIPE
admin-c:        MH375-RIPE
admin-c:        SK2374-RIPE
```

```
% Information related to '95.217.0.0/16AS24940'

route:          95.217.0.0/16
org:            ORG-HOA1-RIPE
descr:          HETZNER-DC
origin:         AS24940
mnt-by:         HOS-GUN
created:        2017-08-12T12:01:47Z
last-modified:  2018-01-10T08:49:24Z
source:         RIPE

organisation:   ORG-HOA1-RIPE
org-name:       Hetzner Online GmbH
country:        DE
org-type:       LIR
address:        Industriestrasse 25
address:        D-91710
address:        Gunzenhausen
address:        GERMANY
phone:          +49 9831 5050
fax-no:         +49 9831 5053
e-mail:         info@hetzner.de
admin-c:        MF1400-RIPE
admin-c:        GM834-RIPE
admin-c:        HOAC1-RIPE
admin-c:        MH375-RIPE
admin-c:        SK2374-RIPE
admin-c:        SK8441-RIPE
abuse-c:        HOAC1-RIPE
mnt-ref:        RIPE-NCC-HM-MNT
mnt-ref:        HOS-GUN
mnt-by:         RIPE-NCC-HM-MNT
mnt-by:         HOS-GUN
created:        2004-04-17T11:07:58Z
last-modified:  2022-11-22T18:32:44Z
source:         RIPE
```

```
role:           Hetzner Online GmbH - Contact Role
address:        Hetzner Online GmbH
address:        Industriestrasse 25
address:        D-91710 Gunzenhausen
address:        Germany
phone:          +49 9831 505-0
fax-no:         +49 9831 505-3
e-mail:         ripe@hetzner.com
abuse-mailbox:  abuse@hetzner.com
remarks:        ************************************************
remarks:        * For spam/abuse/security issues please contact *
remarks:        * abuse@hetzner.com, or fill out the form at *
remarks:        * abuse.hetzner.com, thank you. *
remarks:        ************************************************
remarks:
remarks:        ************************************************
remarks:        * Any questions on Peering please send to *
remarks:        * peering@hetzner.com *
remarks:        ************************************************
org:            ORG-HOA1-RIPE
admin-c:        MH375-RIPE
tech-c:         GM834-RIPE
tech-c:         SK2374-RIPE
tech-c:         MF1400-RIPE
tech-c:         SK8441-RIPE
tech-c:         DD15478-RIPE
nic-hdl:        HOAC1-RIPE
notify:         ripe-mntner@hetzner.com
mnt-by:         HOS-GUN
created:        2004-08-12T09:40:20Z
last-modified:  2022-11-22T18:33:55Z
source:         RIPE
```

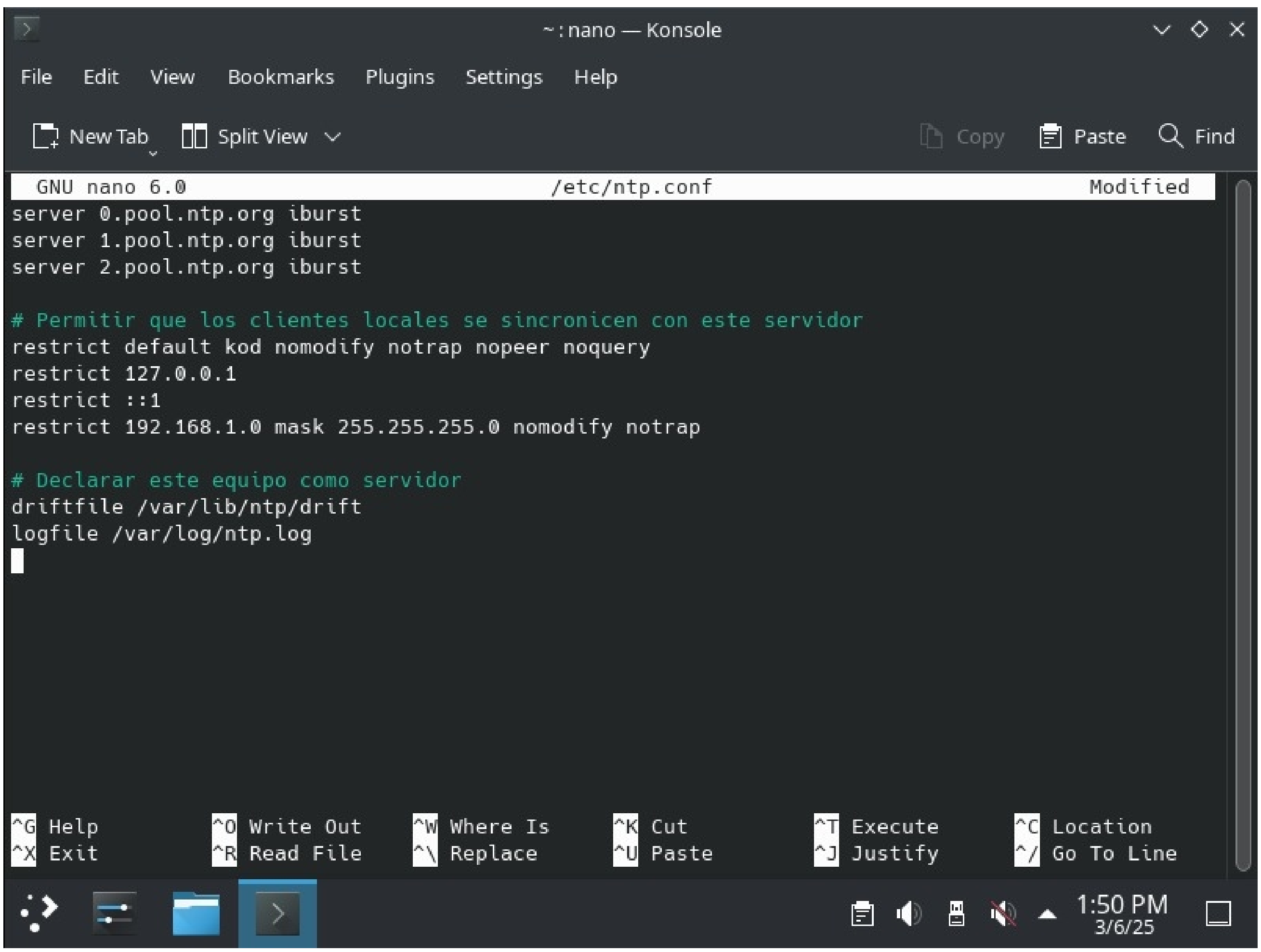
NTP SERVER

When there are millions of machines in the world, it is important that they are somehow synchronized for certain tasks. In this case, global servers running the NTP protocol are used so that all machines that connect to that server can synchronize their clocks.

In this part, we are asked to set up a server and clients, using the protocol and virtual machines we already had. To do this, we start by installing NTP on Slackware with `***slackpkg install ntp***` and NetBSD with `***pkgin install ntp***`. In the case of Windows, it has default access to the protocol through the console.

Now we are going to load the necessary configurations so that the system can start correctly. These configurations are simply to be able to connect to a root server with NTP and connect the other machines that will use the service that we are going to start.

Slackware:



NetBSD:

```
server 0.pool.ntp.org iburst
server 1.pool.ntp.org iburst
server 2.pool.ntp.org iburst

restrict default kod nomodify notrap nopeer noquery
restrict 127.0.0.1
restrict ::1
restrict 192.168.1.0 mask 255.255.255.0 nomodify notrap

driftfile /var/lib/ntp/drift
logfile /var/log/ntp.log







# $NetBSD: ntp.conf,v 1.20.16.1 2020/10/08 16:53:57 martin Exp $
#
#
# NetBSD default Network Time Protocol (NTP) configuration file for ntpd

"/etc/ntp.conf" 156L, 5617B written                          10,1          Top
```

Finally, we are going to start the service using commands for each operating system.

- Slackware: `*chmod +x /etc/rc.d/rc.ntpd*` y `*/etc/rc.d/rc.ntpd start*`.
- NetBSD: `*echo ntpd=YES >> /etc/rc.conf* ` y `*service ntpd start*`.
- Windows Server: `*w32tm /config /manualpeerlist:"192.168.1.100" /syncfromflags:manual /update*` y `*w32tm /resync*`.

Once we have everything ready, we are going to divide this process into 2, since the guide asks us to simulate 2 work groups.

Team 1:

Initialize the server and set the IP in the client configurations. Then, we see that when executing the command `***ntpq -p***` in Slackware and the command `***w32tm /query /status***` on Windows, BSD will be shown as connected, indicating that it is well configured and functional.

```
root@root:~# ntpq -p
     remote           refid      st t when poll reach   delay   offset  jitter
==============================================================================
+0.cl.ntp.edgeun 132.163.96.1     2 u   30   64    7   70.109   +3.151   2.137
+0.co.ntp.edgeun 169.229.128.134  2 u    1   64    3    2.663   +3.306   2.162
*1.co.ntp.edgeun 132.163.96.3     2 u   66   64    3    3.329   +2.558   2.203
root@root:~# nano etc/ntp.conf
root@root:~# nano /etc/ntp.conf
root@root:~# /etc/rc.d/rc.ntpd restart
Stopping NTP daemon...
Starting NTP daemon:  /usr/sbin/ntpd -g -u ntp:ntp
root@root:~# ntpq -p
     remote           refid      st t when poll reach   delay   offset  jitter
==============================================================================
 10.2.77.29      .INIT.          16 u   57   64    0    0.000   +0.000   0.000
*0.cl.ntp.edgeun 132.163.96.1     2 u   57   64    3   69.430   +1.396   2.915
+0.co.ntp.edgeun 169.229.128.142  2 u   54   64    3    3.141   +1.516   3.614
+1.co.ntp.edgeun 129.6.15.27      2 u   54   64    3    3.253   +1.271   3.390
```

```
C:\Windows\system32>w32tm /stripchart /computer:10.2.77.29 /dataonly /samples:5
Tracking 10.2.77.29 [10.2.77.29:123].
Collecting 5 samples.
The current time is 3/6/2025 2:43:50 PM.
14:43:50, -01.0897205s
14:43:52, -01.0892714s
14:43:54, -01.0888299s
14:43:56, -01.0883915s
14:43:58, -01.0879519s
```

```
C:\Windows\system32>w32tm /query /status
Leap Indicator: 0(no warning)
Stratum: 4 (secondary reference - syncd by (S)NTP)
Precision: -23 (119.209ns per tick)
Root Delay: 0.0856230s
Root Dispersion: 7.8035656s
ReferenceId: 0x0A024D1C (source IP:  10.2.77.28)
Last Successful Sync Time: 3/6/2025 2:18:43 PM
Source: 10.2.77.28
Poll Interval: 6 (64s)


C:\Windows\system32>_
```

Team 2:

Initialize the server and set the IP in the client configurations. Then, we see that when executing the command `***ntpq -p***` in NetBSD and the command `***w32tm /query /status***` on Windows, Slackware appears connected, indicating that it is well configured and functional.

```
net# ntpq -p
     remote           refid      st t when poll reach   delay   offset  jitter
==============================================================================
 10.2.77.28      66.90.89.16      3 u    6   64    1    0.947  -368.80   0.000
 2.netbsd.pool.n .POOL.          16 p    -   64    0    0.000    0.000   0.000
 cronos.unad.edu 200.25.3.11      3 u    2   64    1    4.743  -368.74   0.471
 1.co.ntp.edgeun .STEP.          16 u  935   64    0    0.000    0.000   0.000
 2803:bc40:8160: .STEP.          16 u    -   64    0    0.000    0.000   0.000
net# 
```

```
C:\Windows\system32>w32tm /query /status
Leap Indicator: 0(no warning)
Stratum: 4 (secondary reference - syncd by (S)NTP)
Precision: -23 (119.209ns per tick)
Root Delay: 0.0856230s
Root Dispersion: 7.8035656s
ReferenceId: 0x0A024D1C (source IP:  10.2.77.28)
Last Successful Sync Time: 3/6/2025 2:18:43 PM
Source: 10.2.77.28
Poll Interval: 6 (64s)


C:\Windows\system32>_
```

STRUCTURED CABLING & CABLE CONSTRUCTION

Now that we know how the transport and application layers work, we are going to punch our own Ethernet cables to be able to experiment and see the behavior of the network using the hardware provided for this specific practice. So that we understand the types of connections and their usefulness for the physical layer and data transport.

PATCH CORD CONSTRUCTION

Following the teacher's explanation of the punching procedure, we were requested to do a straight punch and a cross punch. But first we have to understand what each one is for and how they work.

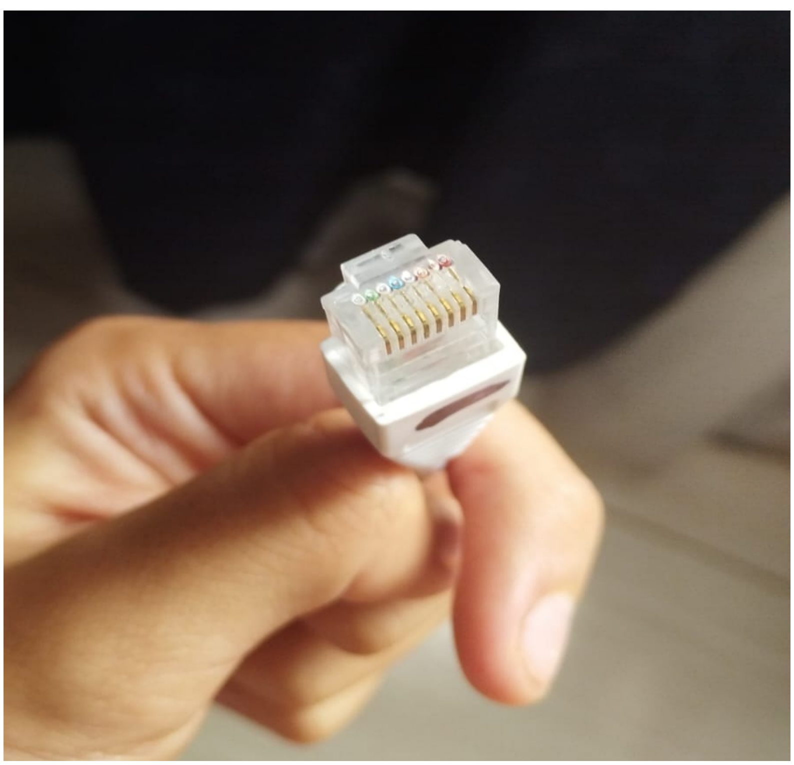
- The straight-through network cable does not change its direction. Both ends use the same wiring standard: T-568A or T-568B. Therefore, both ends of the straight-through cable have the same colored wire arrangement. Thus, Pin 1 on connector A goes to Pin 1 on connector B, Pin 2 to Pin 2, etc.
- A crossover cable, as the name suggests, crosses or changes direction from one end to the other. Unlike a straight-through cable, a crossover cable uses different wiring standards at each end: one is the T568A standard and the other is the T568B standard.

Generally, a crossover cable is used to connect two devices of the same type, such as a PC to a PC or a switch to another switch. On the other hand, a straight-through cable connects two different devices to each other, such as a PC and a switch.

Now we can start the process that begins by cutting the cable to the size we want, then we untangle the cables and arrange them to start organizing them according to each type of connection we will make, then we place the cover and the connector so that the cables are at their highest point. Finally, we use the crimping tool so that the connector is tight, and the cables make contact with the copper.

- Andrés:

- David's:

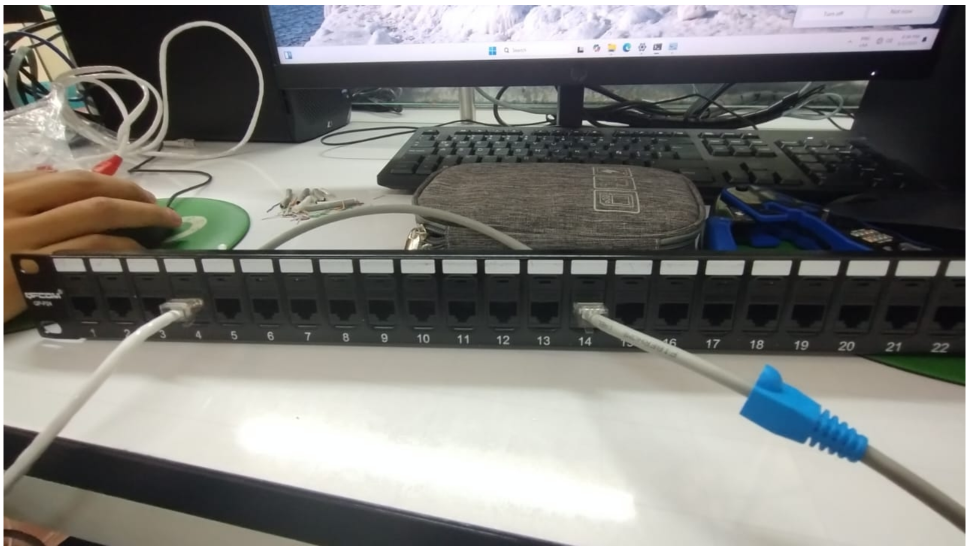


After each cable was done, we proceed to use a device that allows us to perform a test, so we can know if it is functional or if we need to punch the cable again. In our case, it worked the first time, the direct ones indicated pairs of illuminated bulbs and the crossed ones if they complied with the sequences set by the device.

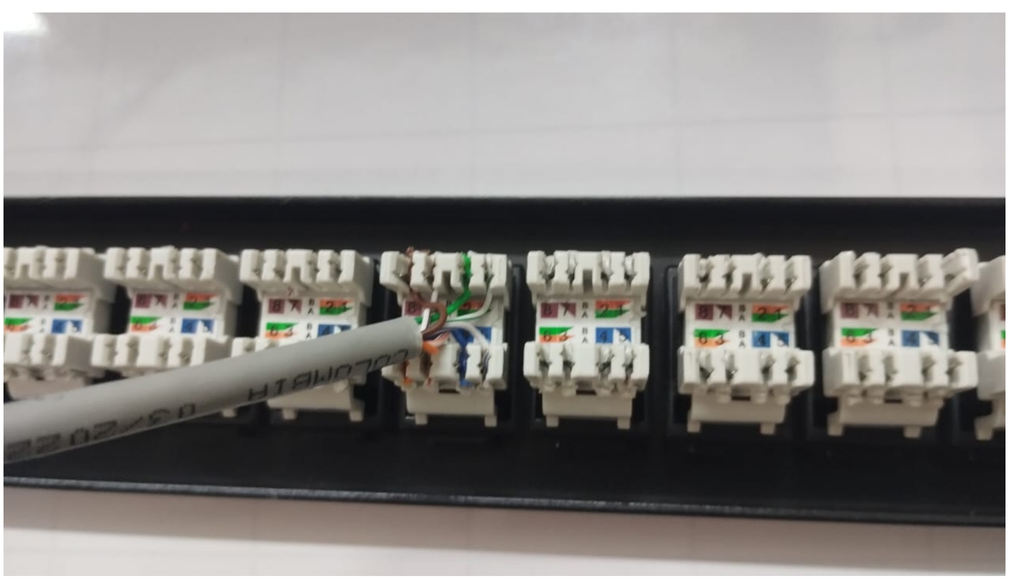
This means that our cables are ready to be tested in practice, meaning they can be used to make network connections to any compatible device.

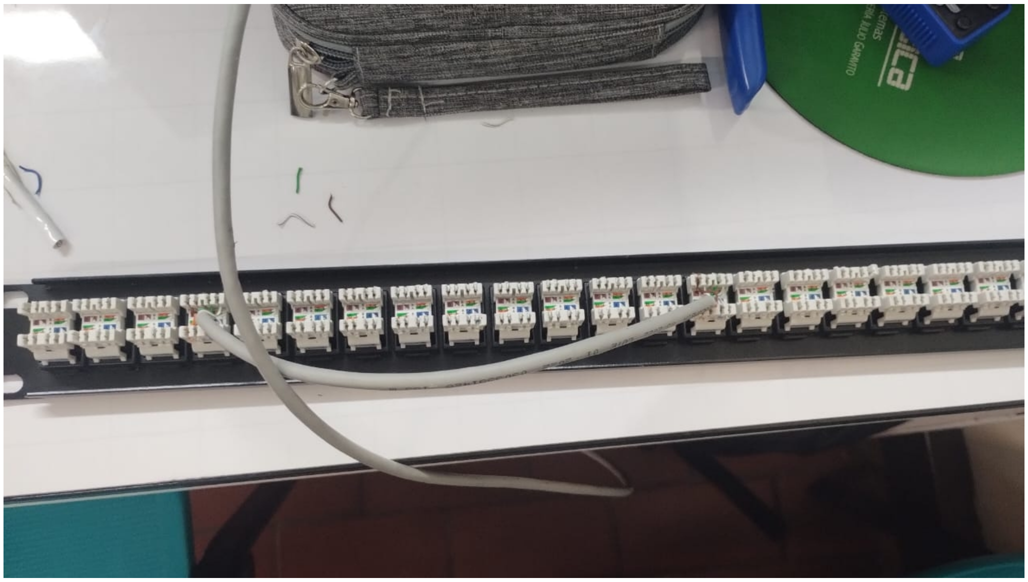
PATCH PANEL CRIMPING

With the faceplate that they gave us in the lab, we are going to check that our cables do work. In addition, we can try another way of crimping cables with a hammer crimping tool, since the plate did not have pre-installed cables. Following the same process as before, we were able to patch with the tool and the guide on the plate itself.

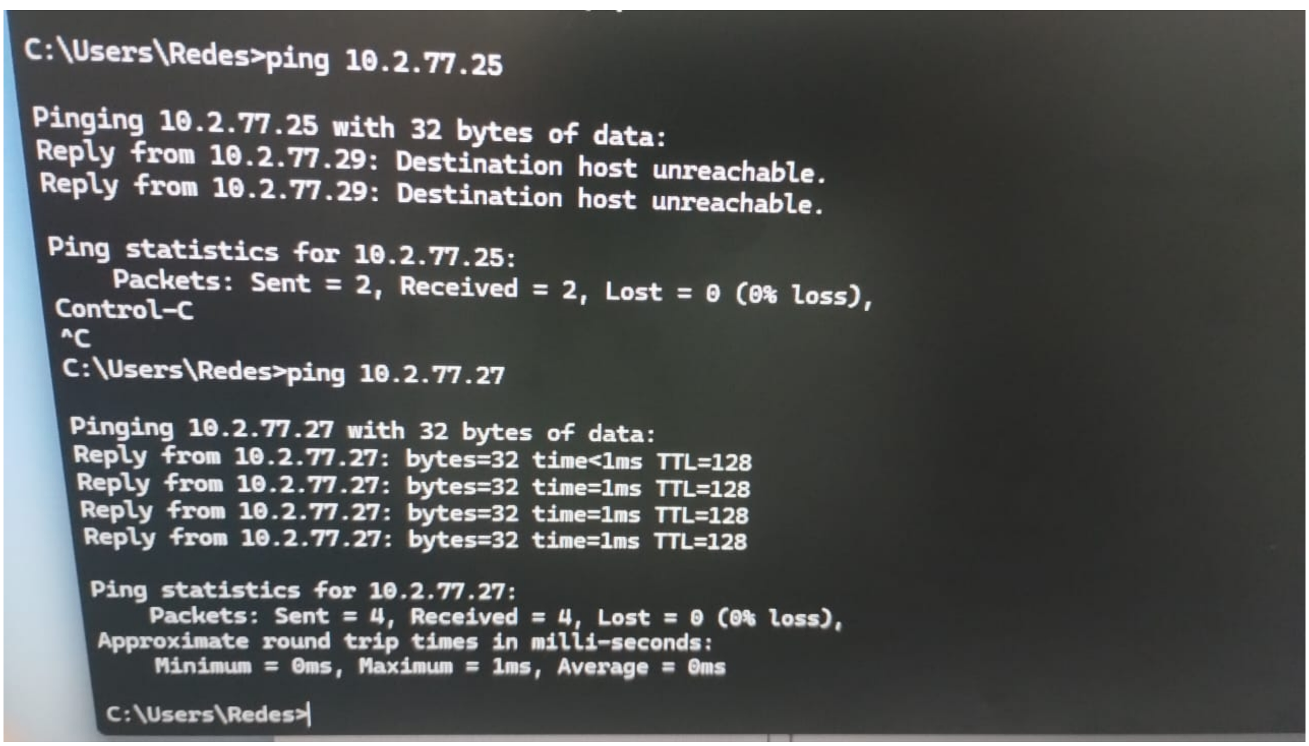


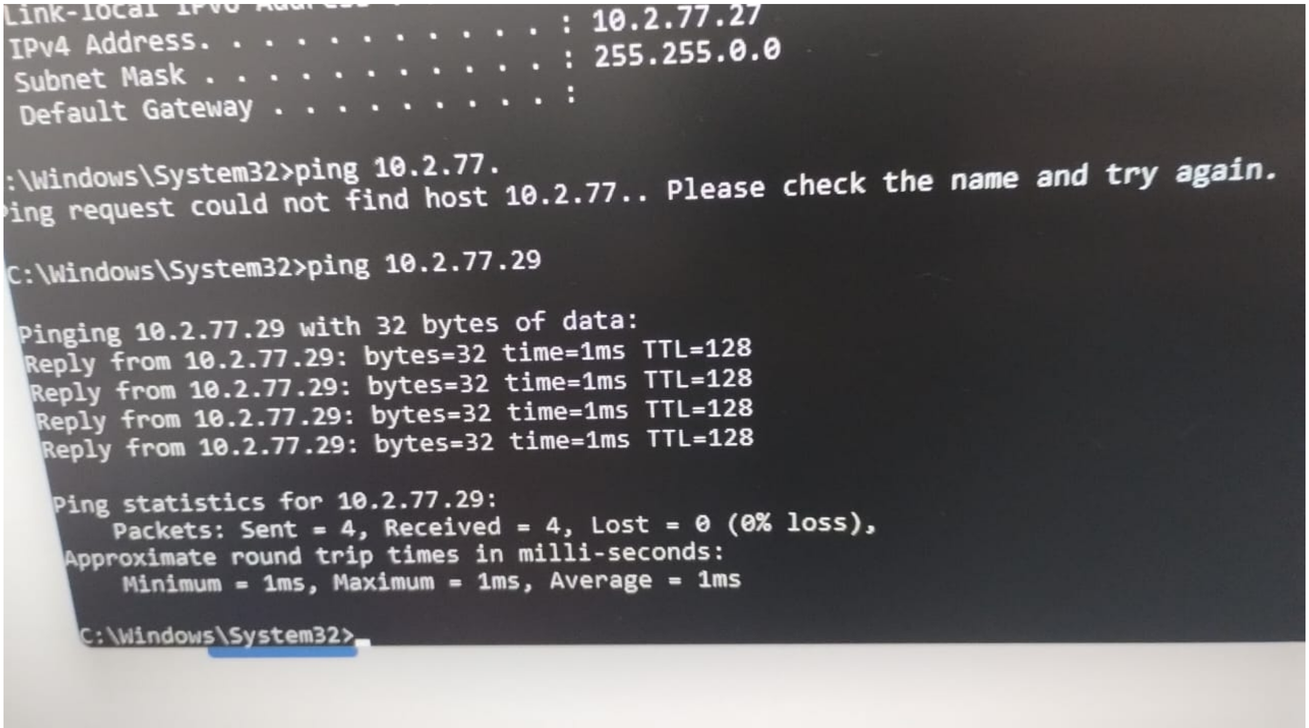
Now we are going to connect the board to two computers. Since these two are connected by the board and the cables, we can know if there is communication between the two devices by using the terminal and a ping to the IP of the other machine. Obviously, a static IP was set to be able to carry out this procedure.





As you can see in the next images, it was possible to ping between the two machines, which proves that the cables and the connections made were done perfectly. In addition, we had the endorsement of the teacher, who validated that we did the complete practice.





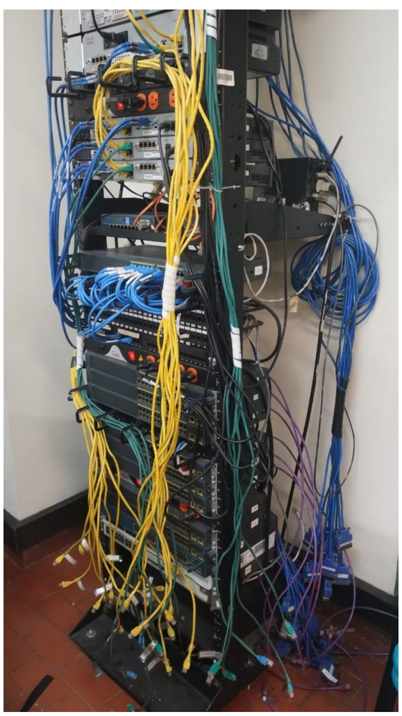
KNOWLEDGE OF THE UNIVERSITY'S CABLING

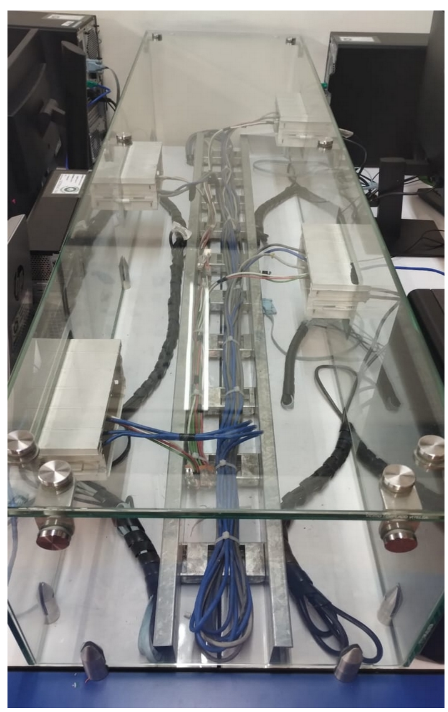
In this last part, we had to go to building I to observe how the components were distributed and their structuring throughout the building in terms of network elements.

The objective of the visit is to learn about and analyze the two types of cabling in structures. When the cabling runs inside the structure and is not easily visible, it is known as vertical cabling. In the case of horizontal cabling, the cables can be seen, and the connections between the different devices that interact with the network can be appreciated.

In this building we will be able to see the vertical cabling, as it is an open building where at first glance you can see the organisation of cables throughout the building.

In this same structure, horizontal cabling elements can be found just like in building B. Elements such as routes, clusters, racks, etc.





CONCLUTIONS

In this lab, we successfully achieved the proposed objectives, exploring both application layer protocols and the physical network infrastructure. Through Cisco Packet Tracer, we configured and analyzed essential services such as DNS, HTTP, FTP, and email, ensuring their proper functionality through connectivity tests and packet analysis.

On the physical side, we applied the structured cabling standard by constructing and verifying network cables, performing patch panel crimping, and testing continuity with specialized tools. These activities helped us understand the importance of proper and organized installation to ensure an efficient and scalable network infrastructure.

Additionally, with Wireshark, we analyzed real network traffic, identifying communication between protocols and evaluating the behavior of services like DHCP and HTTP, strengthening our ability to diagnose and optimize networks. We also performed NTP tests to synchronize time across multiple systems, ensuring temporal consistency in the infrastructure.

In conclusion, this lab provided us with a comprehensive experience, covering both the logical configuration of services and the physical implementation of the network, reinforcing our understanding of the importance of a well-structured IT infrastructure. Through teamwork and the application of theoretical knowledge in a practical environment, we strengthened key skills for network management and administration in real-world scenarios.

BIBLIOGRAFY

DNS Checker. (2025, marzo 6). GeeksforGeeks. https://www.geeksforgeeks.org/dns-checker/

DNS Checker - DNS check propagation tool. (s/f). DNS Checker. Recuperado el 8 de marzo de 2025, de https://dnschecker.org/

FS. (s/f). *¿Cuál es la diferencia entre cable de red directo y cable de red cruzado?* FS.com. Recuperado el 8 de marzo de 2025, de https://www.fs.com/es/blog/patch-cable-vs-crossover-cable-what-is-the-difference-4767.html

Gridelli, S. (2023, junio 13). *How to use telnet to test connectivity to TCP ports*. NetBeez; NetBeez, Inc. https://netbeez.net/blog/telnet-to-test-connectivity-to-tcp/

Matias, G. (2024, agosto 24). Entendiendo el Cableado Horizontal y Vertical, Cableado Estructurado. *Estudios de conexión*. https://estudiosdeconexion.com/telecomunicaciones/entendiendo-el-cableado-horizontal-y-vertical-en-el-contexto-del-cableado-estructurado-estudios-de-conexion-como-proveedor-destacado/

Network time protocol (NTP) tutorial. (s/f). 9tut.com. Recuperado el 8 de marzo de 2025, de https://www.9tut.com/network-time-protocol-ntp-tutorial

Rocío, G. R. (2020, febrero 9). *Tipos de cables de red Ethernet*. ADSLZone. https://www.adslzone.net/reportajes/tecnologia/cables-ethernet-red/

Transport and application layers - OMSCS notes. (s/f). Omscs-notes.com. Recuperado el 8 de marzo de 2025, de https://www.omscs-notes.com/computer-networks/transport-and-application-layers/

Zhang, C. (2018, septiembre 28). *How to use Cat6a patch panel for network cabling?* Medium. https://medium.com/@Camilla.Zhang/how-to-use-cat6a-patch-panel-for-network-cabling-8703fc1ab01